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Using a pseudonym for access authentication over non-3GPP access

Abstract

Apparatuses, methods, and systems are disclosed for using a pseudonym for access authentication over non-3GPP access. One apparatus includes a processor and a transceiver that communicates with a mobile communication network using a 3GPP access network and a non-3GPP access network. The processor sends a registration message to a first network function in the mobile communication network via the 3GPP access network, the first authentication message comprising a first indicator and a SUCI for the apparatus, wherein the first indicator comprises an indication that the apparatus has the capability for access authentication for non-3GPP access in an EPS. The processor receives a first identity pseudonym for the apparatus in response to the registration message comprising the first indicator and performs access authentication via a non-3GPP access network using the first identify pseudonym.

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Background/Summary

(1) The subject matter disclosed herein relates generally to obtaining a pseudonym to use for access authentication over non-3GPP access technology.

BACKGROUND

(2) The following abbreviations and acronyms are herewith defined, at least some of which are referred to within the following description.

(3) Third Generation Partnership Project (“3GPP”), Fifth-Generation Core network (“5GC”), 5G Mobility Management (“5GMM”) Access and Mobility Management Function (“AMF”), Access Point Name (“APN”), Access Stratum (“AS”), Access Network Information (“ANI”), Application Programming Interface (“API”), Data Network Name (“DNN”), Downlink (“DL”), Enhanced Mobile Broadband (“eMBB”), Evolved Node-B (“eNB”), Evolved Packet Core (“EPC”), Evolved Packet System (“EPS”), Evolved UMTS Terrestrial Radio Access Network (“E-UTRAN”), Home Public Land Mobile Network (“HPLMN”), Home Subscriber Server (“HSS”), International Mobile Subscriber Identity (“IMSI”), IP Multimedia Subsystem (“IMS,” aka “IP Multimedia Core Network Subsystem”), Internet Protocol (“IP”), Long Term Evolution (“LTE”), LTE Advanced (“LTE-A”), Medium Access Control (“MAC”), Mobile Network Operator (“MNO”), Mobility Management Entity (“MME”), Non-Access Stratum (“NAS”), Narrowband (“NB”), Network Function (“NF”), Network Access Identifier (“NAI”), Next Generation (e.g., 5G) Node-B (“gNB”), Next Generation Radio Access Network (“NG-RAN”), New Radio (“NR”), Policy Control Function (“PCF”), Packet Data Network (“PDN”), Packet Data Unit (“PDU”), PDN Gateway (“PGW”), Public Land Mobile Network (“PLMN”), Quality of Service (“QoS”), Radio Access Network (“RAN”), Radio Access Technology (“RAT”), Radio Resource Control (“RRC”), Receive (“Rx”), Security Mode Control (“SMC”), Single Network Slice Selection Assistance Information (“S-NSSAI”), Serving Gateway (“SGW”), Session Management Function (“SMF”), Subscription Concealed Identifier (“SUCI”), Subscription Permanent Identifier (“SUPI”), Transmission Control Protocol (“TCP”), Transmit (“Tx”), Trusted Non-3GPP Access Network (“TNAN”), Trusted Non-

3GPP Access Point (“TNAP”), Trusted Non-3GPP Gateway Function (“TNGF”), Unified Data Management (“UDM”), User Entity/Equipment (Mobile Terminal) (“UE”), Uplink (“UL”), User Plane (“UP”), Universal Mobile Telecommunications System (“UMTS”), User Data Management (“UDM”), User Datagram Protocol (“UDP”), User Location Information (“ULI”), User Entity/Equipment (“UE”), UE Parameters Update (“UPU”), Visited Public Land Mobile Network (“VPLMN”), Wireless Local Area Network (“WLAN”), and Worldwide Interoperability for Microwave Access (“WiMAX”).

(4) In certain embodiments, a 5G-capable UE may access an Evolved Packet Core (“EPC”, i.e., 4G core network) via a non-3GPP access network. The identity which is used by a 5G UE for access authentication for non-3GPP access in EPS, as defined in subclause 6 in 3GPP TS 33.402 v16.2.0, is currently sent in clear text. However, this is against the 5G requirement that the identity of a 5G UE must not be exposed.

BRIEF SUMMARY

(5) One method of a remote unit, e.g., a UE, for using a pseudonym for access authentication over non-3GPP access includes sending a registration message to a first network function in the mobile communication network via the 3GPP access network, the first authentication message comprising a first indicator and a SUCI for the UE, wherein the first indicator comprises an indication that the UE has the capability for access authentication for non-3GPP access in an EPS. The method includes receiving a first identity pseudonym for the UE in response to the registration message comprising the first indicator and performs access authentication via a non-3GPP access network using the first identify pseudonym

(6) One method of a first network function, e.g., a UDM, for using a pseudonym for access authentication over non-3GPP access includes receiving a registration request from a remote unit, where the registration request contains a first indicator and a SUCI for the remote unit. The method includes acquiring an identity pseudonym for the remote unit in response to receiving the first indicator and storing a mapping of the identity pseudonym to a subscriber identity of the remote unit. The method includes sending the identity pseudonym to the remote unit, where the mobile communication network uses the identity pseudonym to authenticate the remote unit for non-3GPP access in an EPS.

(7) One method of a second network function, e.g., an AAA server, for using a pseudonym for access authentication over non-3GPP access includes receiving a first authentication message to authenticate a remote unit with the mobile communication network via a non-3GPP access network, the first authentication message comprising a first identity pseudonym for the remote unit. Here, the first identity pseudonym was received by the remote unit during a previous registration with the mobile communication network via a 3GPP access network. The method includes retrieving an authentication vector for the first identity pseudonym, creates a second identity pseudonym for the remote unit, and stores the second identity pseudonym in the mobile communication network. The method includes sending a second authentication message to the remote unit and completes authentication with the remote unit. Here, the second authentication message comprising the second identity pseudonym and a challenge packet derived from the authentication vector.

(8) Another method of a UE for using a pseudonym for access authentication over non-3GPP access includes sending a first authentication message to a second network function to authenticate with the mobile communication network via the non-3GPP access network. Here, the first authentication message comprises a first identity pseudonym received by the UE during a previous registration with the mobile communication network via the 3GPP access network. The method includes receiving a second authentication message from the second network function in response to the first authentication message. Here, the second authentication message comprises a challenge packet and a second identity pseudonym, where the mobile communication network generates the second identity pseudonym in response to the first authentication message comprising a first

identity pseudonym. The method includes completing authentication with the mobile communication network using the challenge packet and locally storing the second identity pseudonym.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) A more particular description of the embodiments briefly described above will be rendered by reference to specific embodiments that are illustrated in the appended drawings. Understanding that these drawings depict only some embodiments and are not therefore to be considered to be limiting of scope, the embodiments will be described and explained with additional specificity and detail through the use of the accompanying drawings, in which:
- (2) FIG. 1 is a diagram illustrating one embodiment of a wireless communication system for using a pseudonym for access authentication over non-3GPP access;
- (3) FIG. 2 is a signal flow diagram illustrating one embodiment of creating an identity pseudonym;
- (4) FIG. 3 is a signal flow diagram illustrating one embodiment of creating an identity pseudonym;
- (5) FIG. 4A is a signal flow diagram illustrating one embodiment of a first solution for TNGF reauthentication;
- (6) FIG. 4B is a continuation of the procedure depicted in FIG. 4A;
- (7) FIG. 4C is a continuation of the procedure depicted in FIG. 4B;
- (8) FIG. 5 is a block diagram illustrating one embodiment of a user equipment apparatus that uses a pseudonym for access authentication over non-3GPP access;
- (9) FIG. 6 is a block diagram illustrating one embodiment of a network equipment apparatus that supports access authentication over non-3GPP access using a pseudonym;
- (10) FIG. 7 is a flow chart diagram illustrating one embodiment of a first method for using a pseudonym for access authentication over non-3GPP access;
- (11) FIG. 8 is a flow chart diagram illustrating one embodiment of a second method for using a pseudonym for access authentication over non-3GPP access;
- (12) FIG. 9 is a flow chart diagram illustrating one embodiment of a third method for using a pseudonym for access authentication over non-3GPP access; and
- (13) FIG. 10 is a flow chart diagram illustrating one embodiment of a fourth method for using a pseudonym for access authentication over non-3GPP access.

DETAILED DESCRIPTION

- (14) As will be appreciated by one skilled in the art, aspects of the embodiments may be embodied as a system, apparatus, method, or program product. Accordingly, embodiments may take the form of an entirely hardware embodiment, an entirely software embodiment (including firmware, resident software, micro-code, etc.) or an embodiment combining software and hardware aspects.
- (15) For example, the disclosed embodiments may be implemented as a hardware circuit comprising custom very-large-scale integration (“VLSI”) circuits or gate arrays, off-the-shelf semiconductors such as logic chips, transistors, or other discrete components. The disclosed embodiments may also be implemented in programmable hardware devices such as field programmable gate arrays, programmable array logic, programmable logic devices, or the like. As another example, the disclosed embodiments may include one or more physical or logical blocks of executable code which may, for instance, be organized as an object, procedure, or function.
- (16) Furthermore, embodiments may take the form of a program product embodied in one or more computer readable storage devices storing machine readable code, computer readable code, and/or program code, referred hereafter as code. The storage devices may be tangible, non-transitory, and/or non-transmission. The storage devices may not embody signals. In a certain embodiment, the storage devices only employ signals for accessing code.

(17) Any combination of one or more computer readable medium may be utilized. The computer readable medium may be a computer readable storage medium. The computer readable storage medium may be a storage device storing the code. The storage device may be, for example, but not limited to, an electronic, magnetic, optical, electromagnetic, infrared, holographic, micromechanical, or semiconductor system, apparatus, or device, or any suitable combination of the foregoing.

(18) More specific examples (a non-exhaustive list) of the storage device would include the following: an electrical connection having one or more wires, a portable computer diskette, a hard disk, a random-access memory (“RAM”), a read-only memory (“ROM”), an erasable programmable read-only memory (“EPROM” or Flash memory), a portable compact disc read-only memory (“CD-ROM”), an optical storage device, a magnetic storage device, or any suitable combination of the foregoing. In the context of this document, a computer readable storage medium may be any tangible medium that can contain, or store, a program for use by or in connection with an instruction execution system, apparatus, or device.

(19) Reference throughout this specification to “one embodiment,” “an embodiment,” or similar language means that a particular feature, structure, or characteristic described in connection with the embodiment is included in at least one embodiment. Thus, appearances of the phrases “in one embodiment,” “in an embodiment,” and similar language throughout this specification may, but do not necessarily, all refer to the same embodiment, but mean “one or more but not all embodiments” unless expressly specified otherwise. The terms “including,” “comprising,” “having,” and variations thereof mean “including but not limited to,” unless expressly specified otherwise. An enumerated listing of items does not imply that any or all of the items are mutually exclusive, unless expressly specified otherwise. The terms “a,” “an,” and “the” also refer to “one or more” unless expressly specified otherwise.

(20) As used herein, a list with a conjunction of “and/or” includes any single item in the list or a combination of items in the list. For example, a list of A, B and/or C includes only A, only B, only C, a combination of A and B, a combination of B and C, a combination of A and C or a combination of A, B and C. As used herein, a list using the terminology “one or more of” includes any single item in the list or a combination of items in the list. For example, one or more of A, B and C includes only A, only B, only C, a combination of A and B, a combination of B and

(21) C, a combination of A and C or a combination of A, B and C. As used herein, a list using the terminology “one of” includes one and only one of any single item in the list. For example, “one of A, B and C” includes only A, only B or only C and excludes combinations of A, B and C. As used herein, “a member selected from the group consisting of A, B, and C,” includes one and only one of A, B, or C, and excludes combinations of A, B, and C. As used herein, “a member selected from the group consisting of A, B, and C and combinations thereof” includes only A, only B, only C, a combination of A and B, a combination of B and C, a combination of A and C or a combination of A, B and C.

(22) Furthermore, the described features, structures, or characteristics of the embodiments may be combined in any suitable manner. In the following description, numerous specific details are provided, such as examples of programming, software modules, user selections, network transactions, database queries, database structures, hardware modules, hardware circuits, hardware chips, etc., to provide a thorough understanding of embodiments. One skilled in the relevant art will recognize, however, that embodiments may be practiced without one or more of the specific details, or with other methods, components, materials, and so forth. In other instances, well-known structures, materials, or operations are not shown or described in detail to avoid obscuring aspects of an embodiment.

(23) Aspects of the embodiments are described below with reference to schematic flowchart diagrams and/or schematic block diagrams of methods, apparatuses, systems, and program products according to embodiments. It will be understood that each block of the schematic flowchart

diagrams and/or schematic block diagrams, and combinations of blocks in the schematic flowchart diagrams and/or schematic block diagrams, can be implemented by code. This code may be provided to a processor of a general-purpose computer, special purpose computer, or other programmable data processing apparatus to produce a machine, such that the instructions, which execute via the processor of the computer or other programmable data processing apparatus, create means for implementing the functions/acts specified in the schematic flowchart diagrams and/or schematic block diagrams.

(24) The code may also be stored in a storage device that can direct a computer, other programmable data processing apparatus, or other devices to function in a particular manner, such that the instructions stored in the storage device produce an article of manufacture including instructions which implement the function/act specified in the schematic flowchart diagrams and/or schematic block diagrams.

(25) The code may also be loaded onto a computer, other programmable data processing apparatus, or other devices to cause a series of operational steps to be performed on the computer, other programmable apparatus, or other devices to produce a computer implemented process such that the code which execute on the computer or other programmable apparatus provide processes for implementing the functions/acts specified in the schematic flowchart diagrams and/or schematic block diagram.

(26) The schematic flowchart diagrams and/or schematic block diagrams in the Figures illustrate the architecture, functionality, and operation of possible implementations of apparatuses, systems, methods, and program products according to various embodiments. In this regard, each block in the schematic flowchart diagrams and/or schematic block diagrams may represent a module, segment, or portion of code, which includes one or more executable instructions of the code for implementing the specified logical function(s).

(27) It should also be noted that, in some alternative implementations, the functions noted in the block may occur out of the order noted in the Figures. For example, two blocks shown in succession may, in fact, be executed substantially concurrently, or the blocks may sometimes be executed in the reverse order, depending upon the functionality involved. Other steps and methods may be conceived that are equivalent in function, logic, or effect to one or more blocks, or portions thereof, of the illustrated Figures.

(28) The description of elements in each figure may refer to elements of proceeding figures. Like numbers refer to like elements in all figures, including alternate embodiments of like elements.

(29) Methods, apparatuses, and systems are disclosed for using a pseudonym for access authentication over non-3GPP access.

(30) Currently the procedure for authentication and key agreement in subclause 6.2 in 3GPP TS 33.402 v16.2.0—which is mandatory for trusted non-3GPP access and optional for untrusted non-3GPP access—foresees that the UE may send its IMSI in clear text, i.e. unencrypted, over that air interface and to the AAA server in the core network. Alternatively, the UE may send (in clear text) a pseudonym allocated to the UE in a previous run of the authentication procedure. A 5G UE is backwards compatible to earlier generations, but the security measures implemented in earlier technologies do not have the same level of security as in 5G, i.e. lower security than in 5G.

(31) The resulting problem is a bidding down attack of a 5G capable UE to make it, expose its secret subscriber identity when the 5G UE is to be authenticated to, e.g., an untrusted non-3GPP access to 5GCN. As discussed above, the 5G UE (likewise 4G UE) may send its secret subscriber identity directly in the first message or as an answer to the identity request message. However, this 4G behavior of a 5G UE may be a violation of the 5G requirement which may require that its subscriber identity (for example, SUPI—which may contain IMSI), may need to be concealed in the first message or as an answer to the identity request message.

(32) Described herein are solutions to the above problem of a 5G UE maintaining the secrecy of the UE permanent subscriber identity, while maintaining backwards compatibility with 4G systems and

procedures. To prevent revealing the UE permanent subscriber identity, a pseudonym is generated by the network, e.g., 5GCN, to use when performing access authentication over non-3GPP access. An example of system architecture for using a pseudonym for access authentication over non-3GPP access is shown in FIG. 1. Examples of procedures to receive an initial pseudonym are shown in FIGS. 2 and 3. An example of a procedure for access authentication using a pseudonym, such as the initial pseudonym, is shown in FIG. 4A-4C.

(33) Note that even if the UE can act as a 4G UE, it is still having the same permanent identity or IMSI as the 5G UE, therefore the 5G requirement of not sending the permanent identity via cleartext should be applied here. Therefore, a dual mode UE (e.g., 5G UE that is also a 4G UE) should always register in a manner that it gets the temporary ID or concealed permanent ID for the 4G registration in case the UE needs to do that registration.

(34) According to a first solution, the 5G UE may register with 5GCN by using 3GPP access. The 5G UE may indicate its capability for access authentication for non-3GPP access in EPS during the registration to the 5GCN by using the 3GPP access. In one embodiment, the access authentication for non-3GPP access in EPS is as defined in subclause 6 in 3GPP TS 33.402. This indication for the capability for access authentication for non-3GPP access in EPS, may be included as a value for the 5GMM capability information element (e.g., defined in 3GPP TS 24.501). The indication for the capability for access authentication for non-3GPP access in EPS may indicate the network that the 5G UE may support the EAP-AKA' authentication method (e.g., defined in IETF RFC 5448).

(35) The indication for the capability for access authentication for non-3GPP access in EPS may hint to the network to provide the UE with information about accessibility to the non-3GPP access networks for authenticating for non-3GPP access in EPS. In one embodiment, this accessibility information may be conveyed to the 5G UE from the PCF via AMF in forms of URSP rules. For the 5G UE to receive such information may be dependent on the user subscription, where if the user subscription allows it, then the PCF transmits the new URSP rules to the 5G UE.

(36) The indication for the capability for access authentication for non-3GPP access in EPS may trigger the UDM to create an initial pseudonym. For this procedure, the UDM may need to obtain the 5G UE's SUPI from the SUCI (using the SIDF functionality offered by the UDM) which may have been used by the 5G UE at the time of registration to the 5GCN. The SIDF may obtain the 5G UE's SUPI from the SUCI by de-concealing SUCI and provide it to the UDM.

(37) In some embodiments, the creation of the initial pseudonym is performed locally at the UDM. The UDM may create the initial pseudonym based on the 5G UE's SUPI where the SUPI type is IMSI), according to techniques discussed herein, and store the created initial pseudonym and the UE identifier (i.e., 5G UE SUPI, 5G UE SUPI of type IMSI, or 5G UE IMSI) at a location of the mobile network where it is accessible by non-3GPP access network. For example, the UDM/UDR may store the 5G UE's SUPI and the created initial pseudonym in an HSS. In other embodiments, the creation of the initial pseudonym is performed by an AAA entity (e.g., a 3GPP AAA server). Here, the UDM sends a request directly to the AAA entity to create the initial pseudonym, the request comprising the 5G UE's IMSI for creation of the initial pseudonym, wherein the AAA entity may create the initial pseudonym based on the 5G UE's IMSI. Note that the AAA entity may be the same or a different AAA entity as the 3GPP AAA server that performs access authentication with the 5G UE for non-3GPP access in EPS.

(38) One option for the generation of the initial pseudonym may be similar to a one time token generation and it can be based on a random number generator/hash function which may create appropriate unique output for the initial pseudonym taking IMSI and any random number/nonce as input.

(39) Alternatively, the UDM (or AAA entity) may generate an initial pseudonym by encrypting the IMSI using any secret key (Home network Private/Secret Key) available in the UDM (or AAA entity). Further, the input to the encryption algorithm at UDM/AAA can be a clear text which is composed of the compressed IMSI concatenated with a random number. The initial pseudonym

may have the format of an IMSI.

(40) The UDM may store the 5G UE's IMSI and the created initial pseudonym in an HSS. The UDR may store the 5G UE's SUPI, 5G UE's SUPI where the SUPI type is IMSI, 5G UE's IMSI and the created initial pseudonym.

(41) The UDM may provide the initial pseudonym in the subscription profile to the AMF which may send the created initial pseudonym to the 5G UE in the registration accept message. Alternatively, the UDM may send the created initial pseudonym to the 5G UE by using the UE parameter update procedure where the UE assigned parameters are updated by the network. The 5G UE may store the initial pseudonym to a non-volatile memory.

(42) The 5G UE may use the initial pseudonym for access authentication for non-3GPP access in EPS, e.g., using the procedure defined in subclause 6 in 3GPP TS 33.402. The 5G UE may use the initial pseudonym for access authentication for non-3GPP access in EPS in order not to reveal its SUPI and/or IMSI. In one embodiment, the SUPI type is IMSI.

(43) In the first solution, the 5G UE may also be a 4G UE or any other generation UE. As an example if the 5G UE is also a 4G UE and is capable to perform the EPC registration, the AAA server may create the initial pseudonym which may be similar to a one time token and can be based on a random number generator which may create appropriate unique output. The initial pseudonym may have the format of an IMSI. The AAA server may store the UE's IMSI and the created initial pseudonym in an HSS.

(44) According to a second solution, after the UDM obtains the 5G UE's SUPI from the SDF, the UDM may send a request comprising the 5G UE's IMSI for creation of the initial pseudonym, towards the HSS. In some embodiments, the creation of the initial pseudonym is performed locally at the HSS. The HSS may create the initial pseudonym based on the 5G UE's permanent identity, according to techniques discussed herein, and store the created initial pseudonym and the UE identifier (i.e., 5G UE SUPI, 5G UE SUPI of type IMSI, or 5G UE IMSI). Alternatively, the HSS may interact with an AAA entity (e.g., a 3GPP AAA server) to create the initial pseudonym based on the 5G UE's IMSI. Here, the HSS sends a request to the AAA entity to create the initial pseudonym, the request comprising the 5G UE's IMSI for creation of the initial pseudonym, wherein the AAA entity may create the initial pseudonym based on the 5G UE's IMSI. Note that the AAA entity may be the same or a different AAA entity as the 3GPP AAA server that performs access authentication with the 5G UE for non-3GPP access in EPS.

(45) One option for the generation of the initial pseudonym may be similar to a one time token generation and it can be based on a random number generator/hash function which may create appropriate unique output for the initial pseudonym taking IMSI and any random number/nonce as input. Alternatively, the creation of the initial pseudonym may include encrypting the IMSI using any secret key (Home network Private/Secret Key) available in the HSS (or AAA). Further, the input to the encryption algorithm at HSS can be a clear text which is composed of the compressed IMSI concatenated with a random number. The initial pseudonym may have the format of an IMSI.

(46) The HSS may store a matching pair of 5G UE's SUPI and the initial pseudonym. Alternatively, the HSS may store a matching pair of 5G UE's IMSI and the initial pseudonym. In one embodiment, the 5G UE's SUPI type is IMSI. The HSS may send the 5G UE's identifier (e.g., 5G UE's SUPI where SUPI type is IMSI, or 5G UE's IMSI) and the initial pseudonym to the UDM/UDR. The UDR may store the 5G UE's SUPI where the SUPI type is IMSI, 5G UE's IMSI and the created initial pseudonym.

(47) In the second solution, the 5G UE may also be a 4G UE or any other generation UE. As an example if the 5G UE is also a 4G UE and is capable to perform the EPC registration, the AAA server may create the initial pseudonym which may be similar to a one time token and can be based on a random number generator which may create appropriate unique output. The initial pseudonym may have the format of an IMSI. The AAA server may store the UE's IMSI and the created initial pseudonym in an HSS.

(48) According to a third solution, it is assumed that the initial pseudonym may be assigned to the 4G UE or the 5G UE (or any other generation UE) by means of configuration which may be but is not limited by using UICC, USIM, SIM or internally in the ME. In all cases, the configuration may be done that the UE or the 5G UE stores the initial pseudonym in a non-volatile memory. The initial pseudonym may at the same time be stored in several UDMs/UDRs, several 3GPP AAA servers, several AAA proxies and several HSS for access authentication for non-3GPP access in EPS as defined in subclause 6 in TS 3GPP 33.402 or any other registration or configuration update.

(49) According to a fourth solution, in order to access authenticate for non-3GPP access in EPS, the 5G UE may use NAI which may be constructed by the initial pseudonym, i.e., pseudonym-NAI, for the EAP-AKA' procedures. In this solution, the 5G UE may have the initial pseudonym, e.g., according to one of the above solutions. At the time of registration with 5GCN by using 3GPP access (e.g., discussed above in the first and second solutions), the 5G UE may receive an indication that the network has the capability for access authentication for non-3GPP access in EPS in the register accept message.

(50) The 5G UE may receive the indication for the network capability for access authentication for non-3GPP access in EPS, as a value for the 5GS network feature support information element (e.g., defined in 3GPP TS 24.501). The network capability for access authentication for non-3GPP access in EPS may hint to the 5G UE that it may require using such an authentication procedure to get access to some, but not all non-3GPP access networks. The network capability for access authentication for non-3GPP access in EPS, may hint the 5G UE that it may require using such an authentication procedure to get access to some services. The 5G UE may connect to the non-3GPP access for the access authentication for non-3GPP access in EPS.

(51) The pseudonym-NAI may be prepended with a special character (e.g., the single digit "7" as described in IETF RFC 5448 and 3GPP TS 23.003) to indicate the EAP-AKA' method for the access authentication for non-3GPP access in EPS. As an example, if the initial pseudonym username is 7y34fa8123 and the IMSI is IMSI is 234150999999999, the NAI may be "7"<7y34fa8123>@nai.epc.mnc150.mcc234.3gppnetwork.org, where the prefix "7" indicates the NAI is a pseudonym-NAI for the EAP-AKA' procedure.

(52) According to 3GPP TS 29.273, the non-3GPP access may use the Diameter Extensible Authentication Protocol (EAP) Application (e.g., specified in IETF RFC 4072) for the SWa interface to send the NAI in the Attribute-Value Pair ("AVP") user-name and EAP payload in EAP-payload AVP towards the possibly existing proxy AAA, to authenticate for non-3GPP access in EPS. The NAI in the AVP user-name and EAP payload in EAP-payload AVP may further be sent towards the 3GPP AAA server by using the Diameter EAP Application (e.g., as specified in IETF RFC 4072, according to 3GPP TS 29.273).

(53) The 3GPP AAA server may extract the pseudonym from the NAI. If the 3GPP AAA can match the pseudonym to the 5G UE's IMSI to determine the 5G UE's identity, then the 3GPP AAA may forward the IMSI to an HSS by using an AVP such as the user-name AVP (described in IETF RFC 4072). However, if the 3GPP AAA cannot match the pseudonym to the 5G UE's IMSI to determine the 5G UE's identity, then the 3GPP AAA may forward the pseudonym to the HSS, which may be a centralized network storage that has stored the matching 5G UE's IMSI for the pseudonym, by using an AVP such as the user-name AVP. Note that in case the 5G UE is in a visited network, then the 3GPP AAA may send the identity of the visited network by using a Diameter AVP such as Visited-Network-Identifier AVP (described in 3GPP TS 29.229).

(54) Upon receipt of the pseudonym, the HSS may match the pseudonym to the 5G UE's IMSI, which may have been stored at the time of creation the initial pseudonym, to determine the 5G UE's identity (e.g., IMSI). The HSS may send the UE's identity such as IMSI to the 3GPP AAA by using an AVP such as the user-name AVP.

(55) The 3GPP AAA may send a request for a set of new authentication vectors by using a Diameter AVP, e.g., SIP-Number-Auth-Items AVP (described in 3GPP TS 29.229). Because the

authentication method is EAP-AKA', the 3GPP AAA may send the identity of the access network by using a Diameter AVP, such as ANID AVP (described in 3GPP TS 29.273). The 3GPP AAA may send the authentication method—which may be EAP-AKA'—by using a Diameter AVP, such as SIP-Authentication-Scheme AVP (described in 3GPP TS 29.229).

(56) The HSS may successfully generate one or more authentication vectors by using the information received from the 3GPP AAA to be sent to the 3GPP AAA. The HSS may send a success result code to the 3GPP AAA by using an AVP, such as the Result-Code AVP (described in IETF RFC 4072). The HSS may send the number of one or more authentications vectors to the 3GPP AAA by using an AVP, such as SIP-Number-Auth-Items (described in 3GPP TS 29.229). The HSS may send the authentication data content to the 3GPP AAA by a grouped AVP, such as SIP-Auth-Data-Item (described in 3GPP TS 29.229).

(57) The grouped AVP may include: the authentication method EAP-AKA' in an AVP, such as SIP-Authentication-Scheme; the authentication challenge RAND; the token AUTN in an AVP, such as SIP-Authenticate (described in 3GPP TS 29.229); the expected response XRES in an AVP such as SIP-Authorization AVP (described in 3GPP TS 29.229); the confidentiality key CK' in an AVP, such as Confidentiality-Key AVP (described in 3GPP TS 29.229); and the integrity key IK' in an AVP, such as Integrity-key AVP (described in 3GPP TS 29.229).

(58) Upon receipt of a successful response and the authentication data from the HSS, the 3GPP AAA server may generate the MSK and EMSK (e.g., as defined in IETF RFC 5448). The 3GPP AAA server may create a new pseudonym to match the 5G UE's IMSI. The 3GPP AAA server may store the new pseudonym locally for future 5G UE registrations to 5GCN or the future 5G UE access authentications for non-3GPP access in EPS. The 3GPP AAA server may store the 5G UE identity and create a binding with the stored pseudonym.

(59) The 3GPP AAA server may forward to the HSS the new pseudonym and the 5G UE's identity, which may be, but is not limited to, the matching 5G UE's IMSI. The HSS may be a centralized network storage that has stored the 5G UE's identity (e.g., the matching 5G UE's IMSI) for the initial pseudonym. If the HSS has already assigned corresponding values for these parameters, the HSS may replace those values with the new values. The HSS may be the centralized network storage if the 5G UE is served by different 3GPP AAA servers.

(60) The 3GPP AAA server may perform authorization challenge by sending an EAP-Request/AKA'-Challenge together with the new created pseudonym towards the 5G UE. The non-3GPP access network may send EAP-Request/AKA'-Challenge which may contain the new created pseudonym, towards the 5G UE. The 5G UE may replace the initial pseudonym with the new pseudonym for the future access authentication for non-3GPP access in EPS, e.g., as defined in subclause 6 in 3GPP TS 33.402 or the any future access authentication.

(61) During this non-3GPP registration, the 4G UE or 5G UE or any other generation UE may perform several periodic re-registrations due to e.g., time expiration or change of location or any other reason. The 3GPP AAA server may therefore create a new pseudonym which replace the old pseudonym within the same non-3GPP registration.

(62) FIG. 1A depicts a wireless communication system **100** for using a pseudonym for access authentication over non-3GPP access, according to embodiments of the disclosure. In one embodiment, the wireless communication system **100** includes at least one remote unit **105**, a 5G-RAN **115**, and a mobile core network **140**. The 5G-RAN **115** and the mobile core network **140** form a mobile communication network. The 5G-RAN **115** may be composed of a 3GPP access network **120** containing at least one cellular base unit **121** and/or a non-3GPP access network **130** containing at least one access point **131**. The remote unit communicates with the 3GPP access network **120** using 3GPP communication links **123** and communicates with the non-3GPP access network **130** using non-3GPP communication links **133**. Even though a specific number of remote units **105**, 3GPP access networks **120**, cellular base units **121**, 3GPP communication links **123**, non-3GPP access networks **130**, access points **131**, non-3GPP communication links **133**, and

mobile core networks **140** are depicted in FIG. 1A, one of skill in the art will recognize that any number of remote units **105**, 3GPP access networks **120**, cellular base units **121**, 3GPP communication links **123**, non-3GPP access networks **130**, access points **131**, non-3GPP communication links **133**, and mobile core networks **140** may be included in the wireless communication system **100**.

(63) In one implementation, the wireless communication system **100** is compliant with the 5G system specified in the 3GPP specifications. More generally, however, the wireless communication system **100** may implement some other open or proprietary communication network, for example, LTE/EPC (referred as ‘4G’) or WiMAX, among other networks. The present disclosure is not intended to be limited to the implementation of any particular wireless communication system architecture or protocol.

(64) In one embodiment, the remote units **105** may include computing devices, such as desktop computers, laptop computers, personal digital assistants (“PDAs”), tablet computers, smart phones, smart televisions (e.g., televisions connected to the Internet), smart appliances (e.g., appliances connected to the Internet), set-top boxes, game consoles, security systems (including security cameras), vehicle on-board computers, network devices (e.g., routers, switches, modems), or the like. In some embodiments, the remote units **105** include wearable devices, such as smart watches, fitness bands, optical head-mounted displays, or the like. Moreover, the remote units **105** may be referred to as UEs, subscriber units, mobiles, mobile stations, users, terminals, mobile terminals, fixed terminals, subscriber stations, user terminals, wireless transmit/receive unit (“WTRU”), a device, or by other terminology used in the art.

(65) The remote units **105** may communicate directly with one or more of the cellular base units **121** in the 3GPP access network **120** via uplink (“UL”) and downlink (“DL”) communication signals. Furthermore, the UL and DL communication signals may be carried over the 3GPP communication links **123**. Similarly, the remote units **105** may communicate with one or more access points **131** in the non-3GPP access network(s) **130** via UL and DL communication signals carried over the non-3GPP communication links **133**. Here, the access networks **120** and **130** are intermediate networks that provide the remote units **105** with access to the mobile core network **140**.

(66) In some embodiments, the remote units **105** communicate with a remote host (e.g., in the data network **150**) via a network connection with the mobile core network **140**. For example, an application **107** (e.g., web browser, media client, telephone/VoIP application) in a remote unit **105** may trigger the remote unit **105** to establish a PDU session (or other data connection) with the mobile core network **140** using the 5G-RAN **115** (e.g., a 3GPP access network **120** and/or a non-3GPP access network **130**). The mobile core network **140** then relays traffic between the remote unit **105** and the remote host using the PDU session. Note that the remote unit **105** may establish one or more PDU sessions (or other data connections) with the mobile core network **140**. A PDU Session may be defined by its parameters: [DNN, Type, SSC mode].

(67) The cellular base units **121** may be distributed over a geographic region. In certain embodiments, a cellular base unit **121** may also be referred to as an access terminal, a base, a base station, a Node-B, an eNB, a gNB, a Home Node-B, a relay node, a device, or by any other terminology used in the art. The cellular base units **121** are generally part of a radio access network (“RAN”), such as the 3GPP access network **120**, that may include one or more controllers communicably coupled to one or more corresponding cellular base units **121**. These and other elements of radio access network are not illustrated but are well known generally by those having ordinary skill in the art. The cellular base units **121** connect to the mobile core network **140** via the 3GPP access network **120**.

(68) The cellular base units **121** may serve a number of remote units **105** within a serving area, for example, a cell or a cell sector, via a 3GPP wireless communication link **123**. The cellular base units **121** may communicate directly with one or more of the remote units **105** via communication

signals. Generally, the cellular base units **121** transmit DL communication signals to serve the remote units **105** in the time, frequency, and/or spatial domain. Furthermore, the DL communication signals may be carried over the 3GPP communication links **123**. The 3GPP communication links **123** may be any suitable carrier in licensed or unlicensed radio spectrum. The 3GPP communication links **123** facilitate communication between one or more of the remote units **105** and/or one or more of the cellular base units **121**.

(69) The non-3GPP access networks **130** may be distributed over a geographic region. Each non-3GPP access network **130** may serve a number of remote units **105** with a serving area. An access point **131** in a non-3GPP access network **130** may communicate directly with one or more remote units **105** by receiving UL communication signals and transmitting DL communication signals to serve the remote units **105** in the time, frequency, and/or spatial domain. Both DL and UL communication signals are carried over the non-3GPP communication links **133**. The 3GPP communication links **123** and non-3GPP communication links **133** may employ different frequencies and/or different communication protocols. In various embodiments, an access point **131** may communicate using unlicensed radio spectrum. The mobile core network **140** may provide services to a remote unit **105** via the non-3GPP access networks **130**, as described in greater detail herein.

(70) In some embodiments, a non-3GPP access network **130** connects to the mobile core network **140** via an interworking entity **135**. The interworking entity **135** provides an interworking between the non-3GPP access network **130** and the mobile core network **140**. The interworking entity **135** supports connectivity via the “N2” and “N3” interfaces. As depicted, both the 3GPP access network **120** and the interworking entity **135** communicate with the AMF **143** using a “N2” interface. The 3GPP access network **120** and interworking entity **135** also communicate with the UPF **141** using a “N3” interface.

(71) In certain embodiments, a non-3GPP access network **130** may be controlled by an operator of the mobile core network **140** and may have direct access to the mobile core network **140**. Such a non-3GPP AN deployment is referred to as a “trusted non-3GPP access network.” A non-3GPP access network **130** is considered as “trusted” when it is operated by the 3GPP operator, or a trusted partner, and supports certain security features, such as strong air-interface encryption. In contrast, a non-3GPP AN deployment that is not controlled by an operator (or trusted partner) of the mobile core network **140**, does not have direct access to the mobile core network **140**, or does not support the certain security features is referred to as a “non-trusted” non-3GPP access network. An interworking entity **135** deployed in a trusted non-3GPP access network **130** may be referred to herein as a Trusted Network Gateway Function (“TNGF”). An interworking entity **135** deployed in a non-trusted non-3GPP access network **130** may be referred to herein as a non-3GPP interworking function (“N3IWF”). While depicted as a part of the non-3GPP access network **130**, in some embodiments the N3IWF may be a part of the mobile core network **140** or may be located in the data network **150**.

(72) In one embodiment, the mobile core network **140** is a 5G core (“5GC”) or the evolved packet core (“EPC”), which may be coupled to a data network **150**, like the Internet and private data networks, among other data networks. A remote unit **105** may have a subscription or other account with the mobile core network **140**. Each mobile core network **140** belongs to a single public land mobile network (“PLMN”). The present disclosure is not intended to be limited to the implementation of any particular wireless communication system architecture or protocol.

(73) The mobile core network **140** includes several network functions (“NFs”). As depicted, the mobile core network **140** includes multiple user plane functions (“UPFs”). Here, the mobile core network **140** includes at least a UPF (“UPF”) **141**.

(74) The mobile core network **140** also includes multiple control plane functions including, but not limited to, an Access and Mobility Management Function (“AMF”) **143**, an Authentication Server Function (“AUSF”) **147**, and a Unified Data Management/Unified Data Repository function

(“UDM/UDR”) **149**. The mobile core network **140** also includes a Home Subscriber Server (“HSS”) **151** and a 3GPP AAA server (“AAA-S”) **153** that provide authentication, authorization, policy control and routing information to access gateways or interworking functions for non-3GPP access. Note that the AAA-S **153** may be consolidated and/or co-located with other network functions in the mobile core network **140**. In certain embodiments, the mobile core network **140** may also include a Session Management Function (“SMF”), a Policy Control Function (“PCF”), a Network Repository Function (“NRF”) (used by the various NFs to discover and communicate with each other over APIs), or other NFs defined for the 5G Core. While the depicted embodiment shows the UDM consolidated with the UDR, in other embodiments the UDM and UDR may be separate entities within the mobile core network **140**.

(75) In various embodiments, the mobile core network **140** supports different types of mobile data connections and different types of network slices, wherein each mobile data connection utilizes a specific network slice. Here, a “network slice” refers to a portion of the mobile core network **140** optimized for a certain traffic type or communication service. A network instance may be identified by a S-NSSAI, while a set of network slices for which the remote unit **105** is authorized to use is identified by NSSAI. In certain embodiments, the various network slices may include separate instances of network functions, such as the SMF and UPF **141**. In some embodiments, the different network slices may share some common network functions, such as the AMF **143**. The different network slices are not shown in FIG. **1A** for ease of illustration, but their support is assumed.

(76) Although specific numbers and types of network functions are depicted in FIG. **1A**, one of skill in the art will recognize that any number and type of network functions may be included in the mobile core network **140**. Moreover, where the mobile core network **140** comprises an EPC, the depicted network functions may be replaced with appropriate EPC entities, such as an MME, S-GW, P-GW, HSS, and the like.

(77) While FIG. **1A** depicts components of a 5G RAN and a 5G core network, the described embodiments for using a pseudonym for access authentication over non-3GPP access apply to other types of communication networks and RATs, including IEEE 802.11 variants, GSM, GPRS, UMTS, LTE variants, CDMA 2000, Bluetooth, ZigBee, Sigfox, and the like. For example, in an 4G/LTE variant involving an EPC, the AMF **143** may be mapped to an MME, the SMF mapped to a control plane portion of a PGW and/or to an MME, the UPF **141** may be mapped to an SGW and a user plane portion of the PGW, the UDM/UDR **149** may be mapped to an HSS, etc.

(78) As depicted, a remote unit **105** (e.g., a UE) may connect to the mobile core network (e.g., to a 5G mobile communication network) via two types of accesses: (1) via 3GPP access network **120** and (2) via a non-3GPP access network **130**. The first type of access (e.g., 3GPP access network **120**) uses a 3GPP-defined type of wireless communication (e.g., NG-RAN) and the second type of access (e.g., non-3GPP access network **130**) uses a non-3GPP-defined type of wireless communication (e.g., WLAN). The 5G-RAN **115** refers to any type of 5G access network that can provide access to the mobile core network **140**, including the 3GPP access network **120** and the non-3GPP access network **130**.

(79) In various embodiments, the remote unit **105** sends a special indicator to the mobile core network indicating its support for the access authentication for non-3GPP access network **130** in EPS. In various embodiments, the remote unit **105** sends this indicator when registering with the mobile communication network **140** via the 3GPP access network **120**. Based on this special indicator, the mobile core network **140** generates and stores an initial pseudonym for the remote unit **105** and sends the same to the remote unit **105**. Thereafter, when performing access authentication for a non-3GPP access network **130** in EPS, the remote unit **105** sends the initial pseudonym so as to conceal its permanent subscriber identity.

(80) FIG. **1B** depicts a block diagram of the architecture of the remote unit **105**, according to embodiments of the disclosure. The remote unit **105** comprises a UE domain **170** which includes the mobile equipment (“ME”) **171** and the Universal Integrated Circuit Card (“UICC”) **181**.

(81) The ME **171** comprises a terminal equipment (“TE”) **173** and a mobile terminal (“MT”) **175**. The TE may be one or more devices offering services to the user **107**. For example, the TE **171** may be the part of the remote unit **105** where the user data is generated in uplink and processed in downlink. The TE **171** may run one or more end-to-end applications with which the user **107** interacts.

(82) The MT **175** performs radio-modem transmission and related functions. For example, the MT **175** performs data and signaling message transfer with a mobile communication network **177** (e.g., the RAN **115** and mobile core network **140**) through the radio interface (i.e., wireless links **123** and/or **133**). Functions performed by the MT **175** include radio transmission and handover, speech encoding and decoding, error detection and correction, signaling and access to the UICC **181**. In some embodiments, the TE **173** and MT **175** may be physically separable devices which are communicatively coupled (e.g., via a terminal adapter) to form the remote unit **105**.

(83) The UICC **181** comprises subscription and/or account information for the remote unit **105**. The UICC **181** securely stores and ensures integrity of personal information, including subscriber identities. Subscriber data stored in the UICC **181** may include a permanent subscriber identity (i.e., IMSI and/or SUPI) and one or more authentication keys. In various embodiments, the UICC **181** comprises one or more subscriber/service identity applications, including but not limited to a Universal Subscriber Identity Module (“USIM”) application **183**, a Subscriber Identity Module (“SIM”) application **185**, and IP multimedia Service Identity Module (“ISIM”) **187**. Note that different types of networks may require different protocols, different authentication procedures, different types of authentication/encryption keys and/or different types of subscriber identities, thus the UICC **181** may include different applications (e.g., USIM **183**, SIM **185**, etc.) to support remote unit **105** access to the different network types. In certain embodiments, the UICC **181** is a removable IC card. In other embodiments, the UICC **181** is an embedded IC. While only a single UICC **181** is depicted in FIG. **1B**, in other embodiments the remote unit **105** may include multiple UICC **181**.

(84) As discussed above, in some embodiments the remote unit **105** may be assigned or preconfigured with an initial pseudonym. For example, the initial pseudonym may be configured using UICC **181**, USIM **183**, SIM **185** or internally in the ME **171**. The initial pseudonym may at the same time be stored in one or more UDMs/UDRs **149**, 3GPP AAA servers **153**, AAA proxies and HSS **151** for access authentication for non-3GPP access in EPS.

(85) FIG. **2** depicts a procedure **200** for creation and storage of an initial pseudonym, according to embodiments of the disclosure. The procedure **200** involves the UE **205** (e.g., one embodiment of the remote unit **105**) that registers with a 5G core network **210** in an HPLMN using a 3GPP access (i.e., 3GPP access network **120**). The 5GC **210** includes an AMF **211** (e.g., one embodiment of the AMF **143**), an AUSF **213** (e.g., one embodiment of the AUSF **147**), and UDM **215** (e.g., one embodiment of the UDM/UDR **149**) that interacts with a HSS **217** (e.g., one embodiment of the HSS **151**).

(86) In the procedure **200**, a 5G UE **205** with capability for access authentication for non-3GPP access in EPS (e.g., as defined in subclause 6 in 3GPP TS 33.402) registers to the 5GC **210** and obtains an initial pseudonym which the 5G UE **205** stores to a non-volatile memory. As used herein, “access authentication for non-3GPP access network in EPS” refers to authentication for the access (i.e., non-3GPP access network **130**) and receiving an IP address. After that the UE will register to the 5GCN network with means of NAS signaling where the UE will be authenticated by the 5GCN. Although the descriptions of FIG. **2** use the terminology “5G UE,” the procedure **200** is not limited to the 5G UE and may be applied to any UE or device.

(87) The procedure **200** begins at Step **1** as the UE **205** registers to the 5GC **210** by using the 3GPP access technology (see block **221**). At the time of registration, the UE **205** may indicate to the network its support for the access authentication for non-3GPP access in EPS. In certain embodiments, this is done by the UE **205** using the 5GMM capability information element, which

may be inserted in the REGISTRATION REQUEST message which may be used by the UE **205** to register to the 5GS. In certain embodiments, the UE **205** only indicates to the network its support for the authentication for non-3GPP access in EPS if it is not preconfigured with a pseudonym. The UE **205** registration may be an initial registration which may be but not limited to the UE **205** may just have initiated the first-time registration to the 5GC **210**. The UE **205** registration may be a mobility and periodic update registration which may be but not limited due to a timer's expiration or change of location.

(88) Upon UE **205**'s registration to the 5GS and indicating to the network its support for the access authentication for non-3GPP access in EPS, the AMF **211** may locate an AUSF **213** and a UDM **215** for the UE **205** authentication. The AMF **211** may initiate to authenticate the UE **205** which may have used its SUCI for this registration.

(89) The AMF **211** may send a Nausf_UEAuthentication_Authenticate request towards the AUSF **213** which may contain the UE **205**'s SUCI. Upon receipt of Nausf_UEAuthentication_Authenticate request containing the UE **205**'s SUCI, the AUSF **213** may send a Nudm_UEAuthentication_Get request containing the UE **205**'s SUCI towards the UDM **215**.

(90) In some embodiments, the UDM **215** creates the initial pseudonym locally, depicted in Option A. In other embodiments, the UDM **215** interacts with the AAA entity **219** to create the initial pseudonym, depicted in Option B.

(91) At step **2** (Option A), the UDM **215** may create an initial pseudonym (see block **223**). The UDM **215** may get the UE **205**'s identity such as SUPI, SUPI with SUPI type IMSI, or IMSI from the SUCI to evaluate if there is already a pseudonym allocated for that identity. The UDM **215** may create the initial pseudonym even if the UE **205** has already a pseudonym from a previous registration, performed by the UE **205**.

(92) At step **3a** (Option B), the UDM **215** sends a create pseudonym request message to the AAA entity **219**, the request containing the UE **205**'s permanent identity (see messaging **225**). The AAA entity **219** creates the initial pseudonym and sends it to the UDM **215** (see messaging **227**). Note that the AAA entity **219** may be the same or a different AAA entity as the 3GPP AAA server that performs access authentication with the 5G UE for non-3GPP access in EPS.

(93) The initial pseudonym may be created but not limited, based on the UE **205**'s SUPI, UE **205**'s SUPI of type IMSI, or UE **205**'s IMSI or based on a random number generator which creates appropriate unique output. Alternatively, the initial pseudonym may be created by the UDM **215** or AAA entity **219** by encrypting the IMSI by its private/secret key.

(94) The initial pseudonym may have the format of an IMSI. The created initial pseudonym may be locally stored along with the UE **205**'s identity in the UDM **215** (e.g., where the UDM **215** is combined or co-located with the UDR).

(95) At step **4**, the UDM **215** stores the created initial pseudonym and the UE **205**'s identity in a centralized network storage, which may be an HSS **217**. The mapping of pseudonym to UE identity is stored for future UE **205** registrations to the 5GC **210** or the future UE **205** access authentications for non-3GPP access in EPS. Examples of the UE identity stored with the pseudonym include, but are not limited to, the matching UE **205**'s SUPI, the UE **205**'s SUPI of type IMSI, and the UE **205**'s IMSI. The UDM **215** sends a storage request message (e.g., Store Pseudonym Request) which contains the created initial pseudonym and the UE **205**'s identity (see messaging **229**).

(96) The HSS **217** may be accessible from several UDM **215**s or several 3GPP AAAs for this information. If the HSS **217** has already assigned corresponding values for these parameters, the HSS **217** may replace those values with the recent values it has obtained from the UDM **215**. Note that the stored mapping may be accessible to both 3GPP access networks and non-3GPP access networks.

(97) In certain embodiments of Option B, the AAA entity **219** may send the created initial

pseudonym and UE identity directly to the HSS **217**. In such embodiments, the UDM **215** does not need to send the store pseudonym request to the HSS **217**. Rather, the AAA entity **219** sends the Store Pseudonym request to the HSS **217**.

(98) At step **5**, the HSS **217** may acknowledge to the UDM **215**, the storage of the created initial pseudonym and the UE **205**'s identity (see messaging **229**). In some embodiments, the HSS **217** may deny the storage request, e.g., due to having an existing values already stored. If the HSS **217** denies storing the created initial pseudonym and the UE **205**'s identity, then the HSS **217** may provide the already existing corresponding values to the UDM **215**. Alternatively, where the HSS **217** received the store pseudonym request from the AAA entity **219**, then the HSS **217** may acknowledge to the AAA entity **219** the storage of the created initial pseudonym and the UE **205**'s identity.

(99) At step **6**, the UDM **215** initiates a UE Parameter Update (“UPU”) procedure and sends to the UE **205** the created initial pseudonym (see block **231**). The UDM **215** may send a Nudm_UEAuthentication_Get response containing the initial pseudonym along with authentication parameters to the AUSF **213**. The AUSF **213** may initiate to challenge the UE **205**'s authentication by sending a Nausf_UEAuthentication_Authenticate response towards the UE **205** via the AMF **211**. Once the AUSF **213** receives the UE **205**'s response via the AMF **211** in a Nausf_UEAuthentication_Authenticate request, the AUSF **213** may send a Nausf_UEAuthentication_Authenticate response towards the AMF **211** containing the initial pseudonym along with success result code.

(100) The AMF **211** may send the initial pseudonym towards the UE **205** by using payload container with the payload container type information element of value “UE **205** parameters update transparent container”. In an alternative to step **6**, the UDM **215** may provide the initial pseudonym in the subscription profile to the AMF **211** which may send the created initial pseudonym to the UE **205** in the registration accept message. The payload container may be sent to the UE **205** in a registration accept message which may be sent by the network to the UE **205** or by any message to update the configuration data.

(101) At step **7**, the UE **205** may store the created initial pseudonym or any other pseudonym for the access authentication for non-3GPP access in EPS received from the network, in a non-volatile memory (see block **233**). If the UE **205** has already an existing pseudonym in the non-volatile memory, the UE **205** may replace the already existing pseudonym with the new one which was obtained from the network in message **6**.

(102) FIG. **3** depicts a procedure **300** for creation and storage of an initial pseudonym, according to embodiments of the disclosure. The procedure **300** involves the UE **205**, the 5GC **210**, the AMF **211**, the AUSF **213**, the UDM **215**, and the HSS **217**. The procedure **300** presents an alternative to the procedure **200** discussed above. While the UDM **215** generates the initial pseudonym in the procedure **200**, in the procedure **300** it is the HSS **217** that generates the initial pseudonym.

(103) In the procedure **300**, a 5G UE **205** with capability for access authentication for non-3GPP access in EPS (e.g., as defined in subclause 6 in 3GPP TS 33.402) registers to the 5GC **210** and obtains an initial pseudonym which the 5G UE **205** stores to a non-volatile memory. Although the descriptions of FIG. **3** use the terminology “5G UE,” the procedure **200** is not limited to the 5G UE and may be applied to any UE or device.

(104) The procedure **300** begins at Step **1** as the UE **205** may register to the 5GCN by using the 3GPP access technology (see block **221**). At the time of registration, the UE **205** may indicate to the network its support for the access authentication for non-3GPP access in EPS, as described above with reference to FIG. **2**, Step **1**.

(105) Upon the UE's registration to the 5GS and indicating to the network its support for the access authentication for non-3GPP access in EPS, the AMF **211** may locate an AUSF **213** and a UDM **215** for the UE **205** authentication, as described above with reference to FIG. **2**.

(106) At step **2**, the UDM **215** may send the UE **205**'s SUPI where the SUPI type is IMSI and/or

IMSI to the HSS **217** and may request the HSS **217** to create an initial pseudonym (see messaging **301**). The UDM **215** may get the UE **205**'s identity such as SUPI, SUPI with SUPI type IMSI or IMSI from the SUCI to evaluate if there is already a pseudonym allocated for that identity.

(107) In some embodiments, the HSS **217** creates the initial pseudonym locally, depicted in Option A. In other embodiments, the HSS **217** interacts with the AAA entity **219** to create the initial pseudonym, depicted in Option B.

(108) At step **3** (Option A), the HSS **217** may locally create an initial pseudonym (see block **303**). The created initial pseudonym may be created even if the UE **205** has already a pseudonym from a previous registration, performed by the UE **205**.

(109) At step **4a** (Option B), the HSS **217** sends a create pseudonym request message to the AAA entity **219**, the request containing the UE **205**'s permanent identity (see messaging **305**). At step **4b**, the AAA entity **219** creates the initial pseudonym and sends it to the HSS **217** (see messaging **307**). Note that the AAA entity **219** may be the same or a different AAA entity as the 3GPP AAA server that performs access authentication with the 5G UE for non-3GPP access in EPS.

(110) The initial pseudonym may be created but not limited, based on the UE **205**'s SUPI, the UE **205**'s SUPI of type IMSI, or the UE **205**'s IMSI or based on a random number generator which creates appropriate unique output. Alternatively, the initial pseudonym is created by the HSS **217** or AAA entity **219** encrypting the IMSI by its private/secret key.

(111) The initial pseudonym may have the format of an IMSI. The created initial pseudonym may be locally stored along with the UE **205**'s identity in the HSS **217**. The HSS **217** which may be a centralized network storage, may store the created initial pseudonym and the UE **205**'s identity which may be but not limited to the matching UE **205**'s SUPI, UE **205**'s SUPI of type IMSI, or UE **205**'s IMSI. The HSS **217** may use these stored UE **205**'s identity and the initial pseudonym for future UE registrations to the 5GCN or the future UE access authentications for non-3GPP access in EPS. The HSS **217** may be accessible from several UDM **215**s or several 3GPP AAAs for this information. If the HSS **217** has already assigned corresponding values for these parameters, the HSS **217** may replace those values with the newly created initial pseudonym.

(112) At step **5**, the HSS **217** may send to the UDM **215**/UDR, the initial pseudonym and the UE **205**'s identity (see messaging **309**). The UDR may store the UE **205**'s identity which may be but not limited to the matching UE **205**'s SUPI, UE **205**'s SUPI of type IMSI, or UE **205**'s IMSI and the created initial pseudonym.

(113) At step **6**, the UDM sends the initial pseudonym to the UE inside a Registration Accept message (see block **311**). The UDM **215** may send a Nudm_UEAuthentication_Get response containing the initial pseudonym along with authentication parameters to the AUSF **213**. The AUSF **213** may initiate to challenge the UE **205**'s authentication by sending a Nausf_UEAuthentication_Authenticate response towards the UE **205** via the AMF **211**. Once the AUSF **213** received the UE **205**'s response via the AMF **211** in a Nausf_UEAuthentication_Authenticate request, the AUSF **213** may send a Nausf_UEAuthentication_Authenticate response towards the AMF **211** containing the initial pseudonym along with success result code. The AMF **211** may send the initial pseudonym towards the UE **205** by using payload container with the payload container type information element of value "UE parameters update transparent container". The payload container may be sent to the UE **205** in a registration accept message which may be sent by the network to the UE **205** or by any message to update the configuration data. Alternatively, the UDM **215** may initiate a UE Parameter Update procedure to send the initial pseudonym to the UE **205**.

(114) At step **7**, the UE **205** may store the created initial pseudonym or any other pseudonym for the access authentication for non-3GPP access in EPS received from the network, in a non-volatile memory (see block **233**). If the UE **205** has already an existing pseudonym in the non-volatile memory, then the UE **205** may replace the already existing pseudonym with the new one which was obtained from the network in message **6**.

(115) FIGS. 4A-4C depict a procedure **400** for using a pseudonym for access authentication over non-3GPP access, according to embodiments of the disclosure. The procedure **400** illustrates a first solution for TNGF reauthentication which involves the UE **205**, a non-3GPP access **401** (i.e., one embodiment of the non-3GPP access network **130**), an AAA proxy **403** located in a VPLMN, a 3GPP AAA server **405** located in an EPC in the HPLMN, and the HSS **217** located in the EPC in the HPLMN. In the most typical case, the non-3GPP access network **210** is a WLAN access network complying with the IEEE 802.11 specification.

(116) The procedure **400** represents an example of the procedure where the 5G UE with capability for access authentication for non-3GPP access in EPS (as defined in subclause in TS 3GPP 33.402), performs the access authentication for non-3GPP access in EPS. Although the illustration uses the terminology 5G UE, the procedure is not limited to the 5G UE and may be applied to a UE or device from any generation of the 3GPP technology.

(117) At FIG. 4A, the procedure **400** begins at step **0** as the UE **205** decides to connect to a non-3GPP access network, e.g., using the “access authentication for non-3GPP access in EPS” procedure, as defined in subclause in TS 3GPP 33.402 (see block **407**).

(118) At step **1** the UE **205** first establishes a Layer-2 (L2) connection with an access point in the non-3GPP access **401** (see messaging **409**). In the case of an IEEE 802.11 WLAN, this L2 connection corresponds to an 802.11 Association.

(119) At steps **2-3**, an EAP procedure is initiated by an authenticator in the non-3GPP access **401**. EAP messages are encapsulated into Layer-2 packets, e.g., into IEEE 802.11/802.1x packets. The authenticator in the non-3GPP access **401** sends an EAP Request/Identity message to the UE **205** (see messaging **411**) and the UE **205** sends a Network Access Identifier (“NAI”) as a response in an EAP Response/Identity message (see messaging **413**).

(120) In various embodiments, the UE **205** sends its identity complying with the NAI format “username@realm” (e.g., specified in 3GPP TS 23.003). Note that when the UE **205** is using a pseudonym username instead of the permanent username, the UE **205** selects the realm name portion similarly to how it selects the realm portion when using the permanent username.

(121) In various embodiments, the NAI contains an initial pseudonym, e.g., either allocated to the UE **205** as shown in FIG. 2 or FIG. 3. In other embodiments, the NAI may contain a pseudonym that may have been allocated to the UE **205** in a previous run of the authentication procedure. Further, the NAI may indicate EAP-AKA' as the authentication method for the access authentication for non-3GPP access in EPS.

(122) At step **4**, the non-3GPP access **401** may route the message to the proper 3GPP AAA server **405** based on the realm part of the NAI. Routing the message to the proper 3GPP AAA server **405** may be as described in 3GPP TS 23.003. The routing path may include one or several AAA proxies **403** (see messaging **415**). The access type and the identity of the access network in which the authenticator resides, may be included by the authenticator in the Diameter message. In the case of roaming, the visited network AAA proxy **403** may include the visited network identifier in the same Diameter message. The access network identity for the access network type may be carried in the Diameter message so that UE **205** and HSS **217** use the same access network identities as input for key derivation.

(123) At step **5**, The 3GPP AAA server **405** may receive the EAP Response/Identity message that contains the subscriber identity and the access type over the SWa/SWd interface (see messaging **417**). Upon receipt of the EAP Response/Identity message, the 3GPP AAA server **405** may be able to determine if the UE **205** has already been authenticated by this 3GPP AAA server **405** in a prior EAP-AKA' authentication, if the 3GPP AAA server **405** knows the UE **205**'s IMSI associated with the username of this pseudonym (the format of an pseudonym-NAI is NAI=pseudonym_username@realm). In the case of roaming, the 3GPP AAA server **405** may also receive the visited network identifier in the same Diameter message that carried the EAP Response/Identity message.

(124) At step 6, the 3GPP AAA server **405** may identify the subscriber as a candidate for authentication with EAP-AKA', based on the received identity in the EAP Response/AKA' Identity message. The 3GPP AAA server **405** may extract the pseudonym username from the NAI. If the UE **205** indicates that it supports EAP-AKA', the 3GPP AAA server **405** may check whether it has an unused authentication vector with authentication management field separation bit=1 and the matching access network identifier available for that subscriber. If there is none, a set of new authentication vectors may be retrieved from the HSS **217**.

(125) If the 3GPP AAA server **405** can match the pseudonym username to the UE **205**'s IMSI to determine the UE **205**'s identity to forward to the HSS **217** if requesting authentication vectors. If the 3GPP AAA server **405** cannot match the pseudonym username to the UE **205**'s IMSI to determine the UE **205**'s identity, then the 3GPP AAA server **405** may forward the pseudonym username to the HSS **217** which may have stored the matching UE **205**'s IMSI for the pseudonym username when requesting authentication vectors (see messaging **419**). The 3GPP AAA server **405** may send the identity of the visited network in case the UE **205** is in the visited network and the authentication method (which may be EAP-AKA') to the HSS **217**, if requesting for a set of new authentication vectors.

(126) At step 7a, the HSS **217** may match the pseudonym username to the IMSI, which may have been stored at the time creation on the initial pseudonym, to determine the UE **205**'s identity (see block **421**).

(127) Continuing on FIG. 4B, at step 7b, the HSS **217** may run an AKA algorithm in response to the request from the 3GPP AAA server **405** (see block **423**). Upon receiving from the 3GPP AAA server **405** an indication that the authentication vector is for EAP-AKA', the HSS **217** may generate an authentication vector with authentication management field separation bit=1. The HSS **217** may transform this authentication vector into a new authentication vector by computing CK' and IK' with access network identity may be one of the input parameters.

(128) At step 7c, the HSS **217** may send the transformed authentication vector to the 3GPP AAA server **405** (see messaging **423**). Here, the HSS **217** may send the UE **205**'s identity such as IMSI to the 3GPP AAA.

(129) At step 8, the 3GPP AAA server **405** may create a new pseudonym username to match the UE **205**'s IMSI (see block **427**). The 3GPP AAA server **405** may store the new pseudonym username locally for future UE **205** registrations or future UE **205** access authentications for non-3GPP access in EPS. Additionally, new keying materials MSK and EMSK may be derived from CK' and IK' (e.g., according to IETF RFC 5448).

(130) At step 9, the 3GPP AAA server **405** may forward the new pseudonym username to the HSS **217** which may be a centralized network storage that has stored the matching UE **205**'s IMSI for the pseudonym username (see messaging **429**). If the HSS **217** has already assigned corresponding values for these parameters, the HSS **217** may replace those values with the new values.

(131) At step 10, the 3GPP AAA server **405** may send RAND, AUTN, a message authentication code and the new pseudonym username towards the UE **205** in the access network in an EAP Request/AKA'-Challenge message (see messaging **431**). The 3GPP AAA server **405** may include the access network identity in this message.

(132) At step 11, upon receipt of the new pseudonym username the UE **205** may replace the initial pseudonym username or any other pseudonym username which was used for the EAP-AKA' authentication with the new pseudonym username for the future access authentication for non-3GPP access in EPS (see block **433**). The UE **205** may verify that AUTN is correct and may therefore authenticate the access network by its identity.

(133) At step 12, The UE may send EAP Response/AKA'-Challenge towards the 3GPP AAA server **405** (see messaging **435**).

(134) Continuing on FIG. 4C, at step 13 the 3GPP AAA server **405** may check the received parameters from UE in the EAP-Response/AKA'-Challenge and derive a Master Session Key

(MSK). The 3GPP AAA server **405** may send the EAP Success message to the non-3GPP access **401**, which in turn may forward the EAP Success to the UE **205** (see messaging **437**). Note that the 3GPP AAA server **405** sends the MSK to the non-3GPP access **401**, but the non-3GPP access **401** does not send the MSK to the UE **205**. Rather, the UE **205** derives its own copy of the MSK.

(135) At step **14**, security establishment may be performed between the UE **205** and the non-3GPP access network **401**, e.g., using a key derived from the MSK (see messaging **439**). At step **15**, upon security establishment, the UE **205** may request for an IP request allocation which it may be assigned by the non-3GPP access **401** (see messaging **441**).

(136) Steps **x1**, **x2**, **x3** and **x4** in FIG. **5** refer to the future when the UE **205** is not allocated any IP address and it may connect to a non-3GPP access **401** for the access authentication for non-3GPP access in EPS. Similar to steps **2-4**, above, at steps **x1** and **x2** the UE **205** may receive an EAP Request/Identity message and send an EAP Response/Identity message in response (see messaging **443**, **445**). Here, the UE **205** may send its identity complying with NAI format, where the NAI may contain the new pseudonym. Similar to steps **4-5**, above, at steps **x3** and **x4** the non-3GPP access **401** sends the NAI to the 3GPP AAA server **405**, sent via proxy AAA server **403** when the UE **205** is roaming in the VPLMN (see messaging **447**, **449**).

(137) FIG. **5** depicts one embodiment of a user equipment apparatus **500**, according to embodiments of the disclosure. The user equipment apparatus **500** may be one embodiment of the remote unit **105** and/or the UE **205**. Furthermore, the user equipment apparatus **500** may include a processor **505**, a memory **510**, an input device **515**, an output device **520**, a transceiver **525**. In some embodiments, the input device **515** and the output device **520** are combined into a single device, such as a touch screen. In certain embodiments, the user equipment apparatus **500** does not include any input device **515** and/or output device **520**. Note that the input device **515** and output device **520** may be part of an TE domain of the user equipment apparatus **500**, while the transceiver **525** may be part of a MT domain of the user equipment apparatus **500**.

(138) As depicted, the transceiver **525** includes at least one transmitter **530** and at least one receiver **535**. Here, the transceiver **525** communicates with a mobile core network (e.g., a 5GC) via an access network. Additionally, the transceiver **525** may support at least one network interface **540**. Here, the at least one network interface **540**, e.g., “Uu” interface, facilitates communication with a RAN node. Additionally, the at least one network interface **540** may include an interface used for communications with an AMF, an SMF, and/or a UPF (e.g., N1, N2, and/or N3 interfaces).

(139) The processor **505**, in one embodiment, may include any known controller capable of executing computer-readable instructions and/or capable of performing logical operations. For example, the processor **505** may be a microcontroller, a microprocessor, a central processing unit (“CPU”), a graphics processing unit (“GPU”), an auxiliary processing unit, a field programmable gate array (“FPGA”), or similar programmable controller. In some embodiments, the processor **505** executes instructions stored in the memory **510** to perform the methods and routines described herein. The processor **505** is communicatively coupled to the memory **510**, the input device **515**, the output device **520**, and the transceiver **525**.

(140) In various embodiments, the processor **505** controls the user equipment apparatus **500** to implement the above described UE behaviors. In some embodiments, via the transceiver **525** the processor **505** sends a registration message to a first network function (i.e., AMF) in the mobile communication network via the 3GPP access network, the first authentication message comprising a first indicator and a SUCI for the user equipment apparatus **500**. Here, the first indicator comprises an indication that the user equipment apparatus **500** has the capability for access authentication for non-3GPP access in an EPS. The processor **505** receives (e.g., from the UDM or HSS) a first identity pseudonym (i.e., the initial pseudonym) for the user equipment apparatus **500** in response to the registration message comprising the first indicator and performs access authentication via a non-3GPP access network using the first identify pseudonym.

(141) In some embodiments, the indication that the remote unit has the capability for access

authentication for non-3GPP access in an EPS comprises a 5GMM capability information element. In some embodiments, the processor **505** determines whether an identity pseudonym is preconfigured prior to sending the registration message. In such embodiments, the processor **505** includes the first indicator in response to determining that no identity pseudonym is preconfigured. (142) In one embodiment, a UDM in the mobile communication network generates the first identity pseudonym. In another embodiment, an HSS in the mobile communication network generates the first identity pseudonym. In a further embodiment, an AAA server in the in the mobile communication network. In various embodiments, the mobile communication network locally stores the first identity pseudonym at the HSS.

(143) In various embodiments, the processor **505** performing access authentication via the non-3GPP access network using the first identify pseudonym by sending a first authentication message (e.g., an EAP Response/AKA' Identity message) to a second network function (i.e., 3GPP AAA server) to authenticate with the mobile communication network via the non-3GPP access network. Here, the first authentication message comprising the first identity pseudonym (i.e., a pseudonym-NAI containing the initial pseudonym). The processor **505** receives a second authentication message (e.g., an EAP Request/AKA'-Challenge message) from the second network function (e.g., 3GPP AAA-S) in response to the first authentication message, the second authentication message comprising a challenge packet and a second identity pseudonym (e.g., the AAA-S sends an EAP-Request/AKA'-Challenge together with the new created pseudonym). Here, the mobile communication network generates the second identity pseudonym (i.e., the new pseudonym) in response to the first authentication message comprising a first identity pseudonym (e.g., the initial pseudonym or another previously received pseudonym). The processor **505** completes authentication with the mobile communication network using the challenge packet (i.e., the user equipment apparatus **500** verifies that the AUTN is correct and sends an EAP Response/AKA'-Challenge towards the 3GPP AAA-S) and locally stores the second identity pseudonym (i.e., the new pseudonym).

(144) In some embodiments, the second network function comprises an AAA-S in the mobile communication network. In certain embodiments, the AAA-S generates the second identity pseudonym in response to the first authentication message comprising a first identity pseudonym. In some embodiments, locally storing the second identity pseudonym comprises replacing the first identity pseudonym with the second identity pseudonym. In some embodiments, the first identity pseudonym and second identity pseudonym are one-time tokens for communicating a permanent subscriber identity of the user equipment apparatus **500** in a concealed manner.

(145) The memory **510**, in one embodiment, is a computer readable storage medium. In some embodiments, the memory **510** includes volatile computer storage media. For example, the memory **510** may include a RAM, including dynamic RAM ("DRAM"), synchronous dynamic RAM ("SDRAM"), and/or static RAM ("SRAM"). In some embodiments, the memory **510** includes non-volatile computer storage media. For example, the memory **510** may include a hard disk drive, a flash memory, or any other suitable non-volatile computer storage device. In some embodiments, the memory **510** includes both volatile and non-volatile computer storage media.

(146) In some embodiments, the memory **510** stores data relating to using a pseudonym for access authentication over non-3GPP access, for example storing subscriber identities, identity pseudonyms, security keys, IP addresses, and the like. In certain embodiments, the memory **510** also stores program code and related data, such as an operating system ("OS") or other controller algorithms operating on the user equipment apparatus **500** and one or more software applications.

(147) The input device **515**, in one embodiment, may include any known computer input device including a touch panel, a button, a keyboard, a stylus, a microphone, or the like. In some embodiments, the input device **515** may be integrated with the output device **520**, for example, as a touchscreen or similar touch-sensitive display. In some embodiments, the input device **515** includes a touchscreen such that text may be input using a virtual keyboard displayed on the touchscreen

and/or by handwriting on the touchscreen. In some embodiments, the input device **515** includes two or more different devices, such as a keyboard and a touch panel.

(148) The output device **520**, in one embodiment, may include any known electronically controllable display or display device. The output device **520** may be designed to output visual, audible, and/or haptic signals. In some embodiments, the output device **520** includes an electronic display capable of outputting visual data to a user. For example, the output device **520** may include, but is not limited to, an LCD display, an LED display, an OLED display, a projector, or similar display device capable of outputting images, text, or the like to a user. As another, non-limiting, example, the output device **520** may include a wearable display such as a smart watch, smart glasses, a heads-up display, or the like. Further, the output device **520** may be a component of a smart phone, a personal digital assistant, a television, a table computer, a notebook (laptop) computer, a personal computer, a vehicle dashboard, or the like.

(149) In certain embodiments, the output device **520** includes one or more speakers for producing sound. For example, the output device **520** may produce an audible alert or notification (e.g., a beep or chime). In some embodiments, the output device **520** includes one or more haptic devices for producing vibrations, motion, or other haptic feedback. In some embodiments, all or portions of the output device **520** may be integrated with the input device **515**. For example, the input device **515** and output device **520** may form a touchscreen or similar touch-sensitive display. In other embodiments, all or portions of the output device **520** may be located near the input device **515**.

(150) As discussed above, the transceiver **525** communicates with one or more network functions of a mobile communication network via one or more access networks. The transceiver **525** operates under the control of the processor **505** to transmit messages, data, and other signals and also to receive messages, data, and other signals. For example, the processor **505** may selectively activate the transceiver (or portions thereof) at particular times in order to send and receive messages.

(151) The transceiver **525** may include one or more transmitters **530** and one or more receivers **535**. Although only one transmitter **530** and one receiver **535** are illustrated, the user equipment apparatus **500** may have any suitable number of transmitters **530** and receivers **535**. Further, the transmitter(s) **530** and the receiver(s) **535** may be any suitable type of transmitters and receivers. In one embodiment, the transceiver **525** includes a first transmitter/receiver pair used to communicate with a mobile communication network over licensed radio spectrum and a second transmitter/receiver pair used to communicate with a mobile communication network over unlicensed radio spectrum.

(152) In certain embodiments, the first transmitter/receiver pair used to communicate with a mobile communication network over licensed radio spectrum and the second transmitter/receiver pair used to communicate with a mobile communication network over unlicensed radio spectrum may be combined into a single transceiver unit, for example a single chip performing functions for use with both licensed and unlicensed radio spectrum. In some embodiments, the first transmitter/receiver pair and the second transmitter/receiver pair may share one or more hardware components. For example, certain transceivers **525**, transmitters **530**, and receivers **535** may be implemented as physically separate components that access a shared hardware resource and/or software resource, such as for example, the network interface **540**.

(153) In various embodiments, one or more transmitters **530** and/or one or more receivers **535** may be implemented and/or integrated into a single hardware component, such as a multi-transceiver chip, a system-on-a-chip, an ASIC, or other type of hardware component. In certain embodiments, one or more transmitters **530** and/or one or more receivers **535** may be implemented and/or integrated into a multi-chip module. In some embodiments, other components such as the network interface **540** or other hardware components/circuits may be integrated with any number of transmitters **530** and/or receivers **535** into a single chip. In such embodiment, the transmitters **530** and receivers **535** may be logically configured as a transceiver **525** that uses one more common control signals or as modular transmitters **530** and receivers **535** implemented in the same hardware

chip or in a multi-chip module.

(154) FIG. 6 depicts one embodiment of a network equipment apparatus **600**, according to embodiments of the disclosure. In some embodiments, the network equipment apparatus **600** may be one embodiment of a TNGF (i.e., TNGF1 and/or TNGF2). In other embodiments, the network equipment apparatus **600** may be one embodiment of an AMF. Furthermore, network equipment apparatus **600** may include a processor **605**, a memory **610**, an input device **615**, an output device **620**, a transceiver **625**. In some embodiments, the input device **615** and the output device **620** are combined into a single device, such as a touch screen. In certain embodiments, the network equipment apparatus **600** does not include any input device **615** and/or output device **620**.

(155) As depicted, the transceiver **625** includes at least one transmitter **630** and at least one receiver **635**. Here, the transceiver **625** communicates with one or more remote units **105**. Additionally, the transceiver **625** may support at least one network interface **640**, such as the N1, N2, and N3 interfaces depicted in FIG. 1. In some embodiments, the transceiver **625** supports a first interface for communicating with a RAN node (e.g., cellular base unit **121** and/or non-3GPP access point **131**), a second interface for communicating with one or more network functions in a mobile core network (e.g., a 5GC and/or EPC) and a third interface for communicating with a remote unit (e.g., UE).

(156) The processor **605**, in one embodiment, may include any known controller capable of executing computer-readable instructions and/or capable of performing logical operations. For example, the processor **605** may be a microcontroller, a microprocessor, a central processing unit (“CPU”), a graphics processing unit (“GPU”), an auxiliary processing unit, a field programmable gate array (“FPGA”), or similar programmable controller. In some embodiments, the processor **605** executes instructions stored in the memory **610** to perform the methods and routines described herein. The processor **605** is communicatively coupled to the memory **610**, the input device **615**, the output device **620**, and the transceiver **625**.

(157) In various embodiments, the processor **605** controls the network equipment apparatus **600** to implement the above described UDM behaviors. In some embodiments, via the network interface **640** the processor **605** receives a registration request from a UE (i.e., remote unit **105**), where the registration request contains a first indicator and a SUCI for the UE. The processor **605** acquires an identity pseudonym for the UE in response to receiving the first indicator and stores a mapping of the identity pseudonym to a subscriber identity of the UE. The processor **605** sends the identity pseudonym to the UE, where the mobile communication network uses the identity pseudonym to authenticate the UE for non-3GPP access in an EPS.

(158) In some embodiments, the first indicator comprises an indication that the UE has the capability for access authentication for non-3GPP access in an EPS and a subscriber identity of the UE. In certain embodiments, the indication that the UE has the capability for access authentication for non-3GPP access in an EPS comprises a 5GMM capability information element. In some embodiments, identity pseudonym comprises a one-time token for communicating a permanent subscriber identity of the UE in a concealed manner.

(159) In some embodiments, the identity pseudonym is created using at least one of: a SUPI of the UE and an IMSI. In one embodiment, creating the identity pseudonym comprises encrypting the subscriber identifier. In one embodiment, creating the identity pseudonym comprises using a random number generator and the subscriber identity to generate a unique value. In one embodiment, creating the identity pseudonym comprises using a hash function and the subscriber identity to generate a hash value. In one embodiment, creating the identity pseudonym comprises sending the subscriber identity to an HSS in the mobile communication network and receiving the identity pseudonym from the HSS. In another embodiment, creating the identity pseudonym comprises sending the subscriber identity to a AAA server in the mobile communication network and receiving the identity pseudonym from the AAA server.

(160) In some embodiments, the network equipment apparatus **600** comprises a UDM. In such

embodiments, storing the mapping of the identity pseudonym to a permanent subscriber identity of the UE comprises sending the identity pseudonym to a home subscriber server in the mobile communication network. In one embodiment, sending the identity pseudonym to the UE comprises sending the identity pseudonym in a registration accept message. In another embodiment, sending the identity pseudonym to the UE comprises initiating a UPU procedure, where the identity pseudonym is sent to the UE within UPU data.

(161) In various embodiments, the processor **605** controls the network equipment apparatus **600** to implement the above described AAA-S behaviors. In some embodiments, via the network interface **640** the processor **605** receives a first authentication message to authenticate a remote unit (i.e., a UE) with the mobile communication network via a non-3GPP access network, the first authentication message comprising a first identity pseudonym (e.g., an initial pseudonym) for the remote unit. Here, the first identity pseudonym was received by the remote unit during a previous registration with the mobile communication network via a 3GPP access network. The processor **605** retrieves an authentication vector for the first identity pseudonym, creates a second identity pseudonym (i.e., a new pseudonym) for the remote unit, and stores the second identity pseudonym in the mobile communication network. The processor **605** sends a second authentication message to the remote unit and completes authentication with the remote unit. Here, the second authentication message comprising the second identity pseudonym and a challenge packet derived from the authentication vector.

(162) In some embodiments, the first identity pseudonym and second identity pseudonym are each one-time tokens for communicating a permanent subscriber identity of the remote unit in a concealed manner, where the first and second identity pseudonyms are mapped to a permanent subscriber identity of the remote unit. In some embodiments, retrieving the authentication vector comprises sending the first identity pseudonym to a HSS in the mobile communication network and receiving a permanent subscriber identity of the remote unit, where the HSS stores a mapping of the first identity pseudonym to a permanent subscriber identity of the remote unit. In certain embodiments, storing the second identity pseudonym comprises sending the second identity pseudonym to the HSS.

(163) In some embodiments, the second identity pseudonym is created using at least one of: a subscriber permanent identifier of the remote unit and an international mobile subscriber identity. In one embodiment, creating the second identity pseudonym comprises encrypting a permanent subscriber identity of the remote unit. In one embodiment, creating the second identity pseudonym comprises using a random number generator and a permanent subscriber identity of the remote unit to generate a unique value. In one embodiment, creating the second identity pseudonym comprises using a hash function and a permanent subscriber identity of the remote unit to generate a hash value. In certain embodiments, storing the second identity pseudonym comprises replacing the first identity pseudonym with the second identity pseudonym.

(164) In one embodiment, the processor **605** receives a request to create an identity pseudonym for a remote unit (i.e., UE) from an HSS in the mobile communication network. Here, the request may include the permanent subscriber identity of the remote unit. In response, the processor **605** creates the initial pseudonym and sends the created identity pseudonym to the HSS. In another embodiment, the processor **605** receives a request to create an identity pseudonym for a remote unit (i.e., UE) from an UDM in the mobile communication network. Here, the request may include the permanent subscriber identity of the remote unit. In response, the processor **605** creates the initial pseudonym and sends the created identity pseudonym to the UDM.

(165) The memory **610**, in one embodiment, is a computer readable storage medium. In some embodiments, the memory **610** includes volatile computer storage media. For example, the memory **610** may include a RAM, including dynamic RAM (“DRAM”), synchronous dynamic RAM (“SDRAM”), and/or static RAM (“SRAM”). In some embodiments, the memory **610** includes non-volatile computer storage media. For example, the memory **610** may include a hard

disk drive, a flash memory, or any other suitable non-volatile computer storage device. In some embodiments, the memory **610** includes both volatile and non-volatile computer storage media. (166) In some embodiments, the memory **610** stores data relating to using a pseudonym for access authentication over non-3GPP access, for example storing subscriber identities, identity pseudonyms, security keys, IP addresses, UE contexts, and the like. In certain embodiments, the memory **610** also stores program code and related data, such as an operating system (“OS”) or other controller algorithms operating on the network equipment apparatus **600** and one or more software applications.

(167) The input device **615**, in one embodiment, may include any known computer input device including a touch panel, a button, a keyboard, a stylus, a microphone, or the like. In some embodiments, the input device **615** may be integrated with the output device **620**, for example, as a touchscreen or similar touch-sensitive display. In some embodiments, the input device **615** includes a touchscreen such that text may be input using a virtual keyboard displayed on the touchscreen and/or by handwriting on the touchscreen. In some embodiments, the input device **615** includes two or more different devices, such as a keyboard and a touch panel.

(168) The output device **620**, in one embodiment, may include any known electronically controllable display or display device. The output device **620** may be designed to output visual, audible, and/or haptic signals. In some embodiments, the output device **620** includes an electronic display capable of outputting visual data to a user. For example, the output device **620** may include, but is not limited to, an LCD display, an LED display, an OLED display, a projector, or similar display device capable of outputting images, text, or the like to a user. As another, non-limiting, example, the output device **620** may include a wearable display such as a smart watch, smart glasses, a heads-up display, or the like. Further, the output device **620** may be a component of a smart phone, a personal digital assistant, a television, a table computer, a notebook (laptop) computer, a personal computer, a vehicle dashboard, or the like.

(169) In certain embodiments, the output device **620** includes one or more speakers for producing sound. For example, the output device **620** may produce an audible alert or notification (e.g., a beep or chime). In some embodiments, the output device **620** includes one or more haptic devices for producing vibrations, motion, or other haptic feedback. In some embodiments, all or portions of the output device **620** may be integrated with the input device **615**. For example, the input device **615** and output device **620** may form a touchscreen or similar touch-sensitive display. In other embodiments, all or portions of the output device **620** may be located near the input device **615**.

(170) As discussed above, the transceiver **625** may communicate with one or more remote units and/or with one or more interworking functions that provide access to one or more PLMNs. The transceiver **625** may also communicate with one or more network functions (e.g., in the mobile core network **140**). The transceiver **625** operates under the control of the processor **605** to transmit messages, data, and other signals and also to receive messages, data, and other signals. For example, the processor **605** may selectively activate the transceiver (or portions thereof) at particular times in order to send and receive messages.

(171) The transceiver **625** may include one or more transmitters **630** and one or more receivers **635**. In certain embodiments, the one or more transmitters **630** and/or the one or more receivers **635** may share transceiver hardware and/or circuitry. For example, the one or more transmitters **630** and/or the one or more receivers **635** may share antenna(s), antenna tuner(s), amplifier(s), filter(s), oscillator(s), mixer(s), modulator/demodulator(s), power supply, and the like. In one embodiment, the transceiver **625** implements multiple logical transceivers using different communication protocols or protocol stacks, while using common physical hardware.

(172) FIG. 7 depicts one embodiment of a method **700** for using a pseudonym for access authentication over non-3GPP access, according to embodiments of the disclosure. In various embodiments, the method **700** is performed by a UE, such as the remote unit **105**, the UE **205**, and/or the user equipment apparatus **500**, described above. In some embodiments, the method **700**

is performed by a processor, such as a microcontroller, a microprocessor, a CPU, a GPU, an auxiliary processing unit, a FPGA, or the like.

(173) The method **700** begins and sends **705** sending a registration message to a first network function in a mobile communication network via a 3GPP access network, the first authentication message comprising a first indicator and a SUCI for the apparatus. Here, the first indicator comprises an indication that the apparatus has the capability for access authentication for non-3GPP access in an EPS. The method **700** includes receiving **710** a first identity pseudonym for the apparatus in response to the registration message comprising the first indicator. The method **700** includes performing **715** access authentication via a non-3GPP access network using the first identify pseudonym. The method **700** ends.

(174) FIG. **8** depicts one embodiment of a method **800** for using a pseudonym for access authentication over non-3GPP access, according to embodiments of the disclosure. In various embodiments, the method **800** is performed by a subscription and user data manager, such as the UDM/UDR **149**, the UDM **215**, and/or the network equipment apparatus **600**, described above. In some embodiments, the method **800** is performed by a processor, such as a microcontroller, a microprocessor, a CPU, a GPU, an auxiliary processing unit, a FPGA, or the like.

(175) The method **800** begins and receives **805** a registration request from a remote unit, wherein the registration request contains a first indicator and a SUCI for the apparatus. The method **800** includes acquiring **810** an identity pseudonym for the remote unit in response to receiving the first indicator.

(176) The method **800** includes storing **815** a mapping of the identity pseudonym to a subscriber identity of the remote unit. The method **800** includes sending **820** the identity pseudonym to the remote unit, where the mobile communication network uses the identity pseudonym to authenticate the remote unit for non-3GPP access in an EPS. The method **800** ends.

(177) FIG. **9** depicts one embodiment of a method **900** for using a pseudonym for access authentication over non-3GPP access, according to embodiments of the disclosure. In various embodiments, the method **900** is performed by an authentication server, such as the AUSF **146**, the 3GPP AAA server **405**, and/or the network equipment apparatus **600**, described above. In some embodiments, the method **900** is performed by a processor, such as a microcontroller, a microprocessor, a CPU, a GPU, an auxiliary processing unit, a FPGA, or the like.

(178) The method **900** begins and receives **905** a first authentication message to authenticate a remote unit with the mobile communication network via a non-3GPP access network, the first authentication message comprising a first identity pseudonym for the remote unit. Here, the first identity pseudonym is received by the remote unit during a previous registration with the mobile communication network via a 3GPP access network.

(179) The method **900** includes retrieving **910** an authentication vector for the first identity pseudonym. The method **900** includes creating **915** a second identity pseudonym for the remote unit. The method **900** includes storing **920** the second identity pseudonym in the mobile communication network.

(180) The method **900** includes sending **925** a second authentication message to the remote unit, the second authentication message comprising the second identity pseudonym and a challenge packet derived from the authentication vector. The method **900** includes completing **930** authentication with the remote unit. The method **900** ends.

(181) FIG. **10** depicts one embodiment of a method **1000** for using a pseudonym for access authentication over non-3GPP access, according to embodiments of the disclosure. In various embodiments, the method **1000** is performed by a UE, such as the remote unit **105**, the UE **205**, and/or the user equipment apparatus **500**, described above. In some embodiments, the method **1000** is performed by a processor, such as a microcontroller, a microprocessor, a CPU, a GPU, an auxiliary processing unit, a FPGA, or the like.

(182) The method **1000** begins and sends **1005** a first authentication message to a second network

function to authenticate with the mobile communication network via the non-3GPP access network, the first authentication message comprising a first identity pseudonym received by the apparatus during a previous registration with the mobile communication network via the 3GPP access network.

(183) The method **1000** includes receiving **1010** a second authentication message from the second network function in response to the first authentication message, the second authentication message comprising a challenge packet and a second identity pseudonym, the mobile communication network generating the second identity pseudonym in response to the first authentication message comprising a first identity pseudonym.

(184) The method **1000** includes completing **1015** authentication with the mobile communication network using the challenge packet. The method **1000** includes locally storing **1020** the second identity pseudonym. The method **1000** ends.

(185) Disclosed herein is a first apparatus for using a pseudonym for access authentication over non-3GPP access, according to embodiments of the disclosure. The first apparatus may be implemented by a UE, such as the remote unit **105**, the UE **205**, and/or the user equipment apparatus **500**. The first apparatus includes a processor and a transceiver that communicates with a mobile communication network using a 3GPP access network and a non-3GPP access network. The processor sends a registration message to a first network function in the mobile communication network via the 3GPP access network, the first authentication message comprising a first indicator and a SUCI for the first apparatus, wherein the first indicator comprises an indication that the first apparatus has the capability for access authentication for non-3GPP access in an EPS. The processor receives a first identity pseudonym for the first apparatus in response to the registration message comprising the first indicator and performs access authentication via a non-3GPP access network using the first identify pseudonym.

(186) In some embodiments, the indication that the remote unit has the capability for access authentication for non-3GPP access in an EPS comprises a 5GMM capability information element. In some embodiments, the processor determines whether an identity pseudonym is preconfigured prior to sending the registration message. In such embodiments, the processor includes the first indicator in response to determining that no identity pseudonym is preconfigured.

(187) In some embodiments, the first identity pseudonym is a one-time token for communicating a permanent subscriber identity of the first apparatus in a concealed manner.

(188) In one embodiment, a UDM in the mobile communication network generates the first identity pseudonym. In another embodiment, an HSS in the mobile communication network generates the first identity pseudonym. In a further embodiment, an AAA server in the in the mobile communication network. In various embodiments, the mobile communication network locally stores the first identity pseudonym at least at the UDM and the HSS.

(189) In some embodiments, performing access authentication via the non-3GPP access network using the first identify pseudonym comprises the processor sending a first authentication message to a second network function to authenticate with the mobile communication network via the non-3GPP access network, the first authentication message comprising the first identity pseudonym. The processor receives a second authentication message from the second network function in response to the first authentication message, the second authentication message comprising a challenge packet and a second identity pseudonym. Here, the mobile communication network generates the second identity pseudonym in response to the first authentication message comprising a first identity pseudonym. The processor completes authentication with the mobile communication network using the challenge packet and locally stores the second identity pseudonym.

(190) In some embodiments, the second network function comprises an AAA server in the mobile communication network. In certain embodiments, the AAA server generates the second identity pseudonym in response to the first authentication message comprising a first identity pseudonym. In some embodiments, locally storing the second identity pseudonym comprises replacing the first

identity pseudonym with the second identity pseudonym.

(191) Disclosed herein is a first method for using a pseudonym for access authentication over non-3GPP access, according to embodiments of the disclosure. The first method may be performed by a UE, such as the remote unit **105**, the UE **205**, and/or the user equipment apparatus **500**. The first method includes sending a registration message to a first network function in the mobile communication network via the 3GPP access network, the first authentication message comprising a first indicator and a SUCI for the UE, wherein the first indicator comprises an indication that the UE has the capability for access authentication for non-3GPP access in an EPS. The first method includes receiving a first identity pseudonym for the UE in response to the registration message comprising the first indicator and performs access authentication via a non-3GPP access network using the first identity pseudonym.

(192) In some embodiments, the indication that the remote unit has the capability for access authentication for non-3GPP access in an EPS comprises a 5GMM capability information element. In some embodiments, the first method includes determining whether an identity pseudonym is preconfigured prior to sending the registration message. In such embodiments, the first indicator is included in the registration message in response to determining that no identity pseudonym is preconfigured. In some embodiments, the first identity pseudonym is a one-time token for communicating a permanent subscriber identity of the UE in a concealed manner.

(193) In one embodiment, a UDM in the mobile communication network generates the first identity pseudonym. In another embodiment, an HSS in the mobile communication network generates the first identity pseudonym. In a further embodiment, an AAA server in the mobile communication network. In various embodiments, the mobile communication network locally stores the first identity pseudonym at least at the UDM and the HSS.

(194) In some embodiments, performing access authentication via the non-3GPP access network using the first identity pseudonym comprises sending a first authentication message to a second network function to authenticate with the mobile communication network via the non-3GPP access network, the first authentication message comprising the first identity pseudonym. The access authentication includes receiving a second authentication message from the second network function in response to the first authentication message, the second authentication message comprising a challenge packet and a second identity pseudonym. Here, the mobile communication network generates the second identity pseudonym in response to the first authentication message comprising a first identity pseudonym. The access authentication includes completing authentication with the mobile communication network using the challenge packet and locally storing the second identity pseudonym.

(195) In some embodiments, the second network function comprises an AAA server in the mobile communication network. In certain embodiments, the AAA server generates the second identity pseudonym in response to the first authentication message comprising a first identity pseudonym. In some embodiments, locally storing the second identity pseudonym comprises replacing the first identity pseudonym with the second identity pseudonym.

(196) Disclosed herein is a second apparatus for using a pseudonym for access authentication over non-3GPP access, according to embodiments of the disclosure. The second apparatus may be implemented by a subscription and user data manager, such as the UDM/UDR **149**, the UDM **215**, and/or the network equipment apparatus **600**. The second apparatus includes a processor and a network interface that communicates with a mobile communication network. The processor receives a registration request from a remote unit (i.e., UE), where the registration request contains a first indicator and a SUCI for the remote unit. The processor acquires an identity pseudonym for the remote unit in response to receiving the first indicator and stores a mapping of the identity pseudonym to a subscriber identity of the remote unit. The processor sends the identity pseudonym to the remote unit, where the mobile communication network uses the identity pseudonym to authenticate the remote unit for non-3GPP access in an EPS.

(197) In some embodiments, the first indicator comprises an indication that the remote unit has the capability for access authentication for non-3GPP access in an EPS and a subscriber identity of the remote unit. In certain embodiments, the indication that the remote unit has the capability for access authentication for non-3GPP access in an EPS comprises a 5GMM capability information element. In some embodiments, identity pseudonym comprises a one-time token for communicating a permanent subscriber identity of the remote unit in a concealed manner.

(198) In some embodiments, the identity pseudonym is created using at least one of: a SUPI of the remote unit and an IMSI. In one embodiment, creating the identity pseudonym comprises encrypting the subscriber identifier. In one embodiment, creating the identity pseudonym comprises using a random number generator and the subscriber identity to generate a unique value. In one embodiment, creating the identity pseudonym comprises using a hash function and the subscriber identity to generate a hash value. In one embodiment, creating the identity pseudonym comprises sending the subscriber identity to an HSS in the mobile communication network and receiving the identity pseudonym from the HSS. In another embodiment, creating the identity pseudonym comprises sending the subscriber identity to a AAA server in the mobile communication network and receiving the identity pseudonym from the AAA server.

(199) In some embodiments, the second apparatus comprises a UDM. In such embodiments, storing the mapping of the identity pseudonym to a permanent subscriber identity of the remote unit comprises sending the identity pseudonym to a home subscriber server in the mobile communication network. In one embodiment, sending the identity pseudonym to the remote unit comprises sending the identity pseudonym in a registration accept message. In another embodiment, sending the identity pseudonym to the remote unit comprises initiating a UPU procedure, where the identity pseudonym is sent to the remote unit within UPU data.

(200) Disclosed herein is a second method for using a pseudonym for access authentication over non-3GPP access, according to embodiments of the disclosure. The second method may be performed by a subscription and user data manager, such as the UDM/UDR **149**, the UDM **215**, and/or the network equipment apparatus **600**. The second method includes receiving a registration request from a remote unit, where the registration request contains a first indicator and a SUCI for the remote unit. The second method includes acquiring an identity pseudonym for the remote unit in response to receiving the first indicator and storing a mapping of the identity pseudonym to a subscriber identity of the remote unit. The second method includes sending the identity pseudonym to the remote unit, where the mobile communication network uses the identity pseudonym to authenticate the remote unit for non-3GPP access in an EPS.

(201) In some embodiments, the first indicator comprises an indication that the remote unit has the capability for access authentication for non-3GPP access in an EPS and a subscriber identity of the remote unit. In certain embodiments, the indication that the remote unit has the capability for access authentication for non-3GPP access in an EPS comprises a 5GMM capability information element. In some embodiments, identity pseudonym comprises a one-time token for communicating a permanent subscriber identity of the remote unit in a concealed manner.

(202) In some embodiments, the identity pseudonym is created using at least one of: a SUPI of the remote unit and an IMSI. In one embodiment, creating the identity pseudonym comprises encrypting the subscriber identifier. In one embodiment, creating the identity pseudonym comprises using a random number generator and the subscriber identity to generate a unique value. In one embodiment, creating the identity pseudonym comprises using a hash function and the subscriber identity to generate a hash value. In one embodiment, creating the identity pseudonym comprises sending the subscriber identity to an HSS in the mobile communication network and receiving the identity pseudonym from the HSS. In another embodiment, creating the identity pseudonym comprises sending the subscriber identity to a AAA server in the mobile communication network and receiving the identity pseudonym from the AAA server.

(203) In some embodiments, the subscription and user data manager is a UDM. In such

embodiments, storing the mapping of the identity pseudonym to a permanent subscriber identity of the remote unit comprises sending the identity pseudonym to a home subscriber server in the mobile communication network. In one embodiment, sending the identity pseudonym to the remote unit comprises sending the identity pseudonym in a registration accept message. In another embodiment, sending the identity pseudonym to the remote unit comprises initiating a UPU procedure, where the identity pseudonym is sent to the remote unit within UPU data.

(204) Disclosed herein is a third apparatus for using a pseudonym for access authentication over non-3GPP access, according to embodiments of the disclosure. The third apparatus may be implemented by an authentication server, such as the AUSF **146**, the 3GPP AAA server **405**, and/or the network equipment apparatus **600**. The third apparatus includes a processor and a network interface that communicates with a mobile communication network. The processor receives a first authentication message to authenticate a remote unit with the mobile communication network via a non-3GPP access network, the first authentication message comprising a first identity pseudonym for the remote unit. Here, the first identity pseudonym was received by the remote unit during a previous registration with the mobile communication network via a 3GPP access network. The processor retrieves an authentication vector for the first identity pseudonym, creates a second identity pseudonym for the remote unit, and stores the second identity pseudonym in the mobile communication network. The processor sends a second authentication message to the remote unit and completes authentication with the remote unit. Here, the second authentication message comprising the second identity pseudonym and a challenge packet derived from the authentication vector.

(205) In some embodiments, the first identity pseudonym and second identity pseudonym are each one-time tokens for communicating a permanent subscriber identity of the remote unit in a concealed manner, where the first and second identity pseudonyms are mapped to a permanent subscriber identity of the remote unit. In some embodiments, retrieving the authentication vector comprises sending the first identity pseudonym to a HSS in the mobile communication network and receiving a permanent subscriber identity of the remote unit, where the HSS stores a mapping of the first identity pseudonym to a permanent subscriber identity of the remote unit. In certain embodiments, storing the second identity pseudonym comprises sending the second identity pseudonym to the HSS.

(206) In some embodiments, the second identity pseudonym is created using at least one of: a subscriber permanent identifier of the remote unit and an international mobile subscriber identity. In one embodiment, creating the second identity pseudonym comprises encrypting a permanent subscriber identity of the remote unit. In one embodiment, creating the second identity pseudonym comprises using a random number generator and a permanent subscriber identity of the remote unit to generate a unique value. In one embodiment, creating the second identity pseudonym comprises using a hash function and a permanent subscriber identity of the remote unit to generate a hash value. In certain embodiments, storing the second identity pseudonym comprises replacing the first identity pseudonym with the second identity pseudonym.

(207) In one embodiment, the processor receives a request to create an identity pseudonym for a remote unit (i.e., UE) from an HSS in the mobile communication network. Here, the request may include the permanent subscriber identity of the remote unit. In response, the processor creates the initial pseudonym and sends the created identity pseudonym to the HSS. In another embodiment, the processor receives a request to create an identity pseudonym for a remote unit (i.e., UE) from a UDM in the mobile communication network. Here, the request may include the permanent subscriber identity of the remote unit. In response, the processor creates the initial pseudonym and sends the created identity pseudonym to the UDM.

(208) Disclosed herein is a third method for using a pseudonym for access authentication over non-3GPP access, according to embodiments of the disclosure. The third method may be performed by an authentication server, such as the AUSF **146**, the 3GPP AAA server **405**, and/or the network

equipment apparatus **600**. The third method includes receiving a first authentication message to authenticate a remote unit with the mobile communication network via a non-3GPP access network, the first authentication message comprising a first identity pseudonym for the remote unit. Here, the first identity pseudonym was received by the remote unit during a previous registration with the mobile communication network via a 3GPP access network. The third method includes retrieving an authentication vector for the first identity pseudonym, creates a second identity pseudonym for the remote unit, and stores the second identity pseudonym in the mobile communication network. The third method includes sending a second authentication message to the remote unit and completes authentication with the remote unit. Here, the second authentication message comprising the second identity pseudonym and a challenge packet derived from the authentication vector.

(209) In some embodiments, the first identity pseudonym and second identity pseudonym are each one-time tokens for communicating a permanent subscriber identity of the remote unit in a concealed manner, where the first and second identity pseudonyms are mapped to a permanent subscriber identity of the remote unit. In some embodiments, retrieving the authentication vector comprises sending the first identity pseudonym to a HSS in the mobile communication network and receiving a permanent subscriber identity of the remote unit, where the HSS stores a mapping of the first identity pseudonym to a permanent subscriber identity of the remote unit. In certain embodiments, storing the second identity pseudonym comprises sending the second identity pseudonym to the HSS.

(210) In some embodiments, the second identity pseudonym is created using at least one of: a subscriber permanent identifier of the remote unit and an international mobile subscriber identity. In one embodiment, creating the second identity pseudonym comprises encrypting a permanent subscriber identity of the remote unit. In one embodiment, creating the second identity pseudonym comprises using a random number generator and a permanent subscriber identity of the remote unit to generate a unique value. In one embodiment, creating the second identity pseudonym comprises using a hash function and a permanent subscriber identity of the remote unit to generate a hash value. In certain embodiments, storing the second identity pseudonym comprises replacing the first identity pseudonym with the second identity pseudonym.

(211) In one embodiment, the third method includes receiving a request to create an identity pseudonym for a remote unit (i.e., UE) from an HSS in the mobile communication network. Here, the request may include the permanent subscriber identity of the remote unit. In response, the third method includes creating the initial pseudonym and sending the created identity pseudonym to the HSS. In another embodiment, the third method includes receiving a request to create an identity pseudonym for a remote unit (i.e., UE) from an UDM in the mobile communication network. Here, the request may include the permanent subscriber identity of the remote unit. In response, the third method includes creating the initial pseudonym and sending the created identity pseudonym to the UDM.

(212) Disclosed herein is a fourth apparatus for using a pseudonym for access authentication over non-3GPP access, according to embodiments of the disclosure. The fourth apparatus may be implemented by a UE, such as the remote unit **105**, the UE **205**, and/or the user equipment apparatus **500**. The fourth apparatus includes a processor and a transceiver that communicates with a mobile communication network using a 3GPP access network and a non-3GPP access network. The processor sends a first authentication message to a second network function to authenticate with the mobile communication network via the non-3GPP access network. Here, the first authentication message comprises a first identity pseudonym received by the fourth apparatus during a previous registration with the mobile communication network via the 3GPP access network. The processor receives a second authentication message from the second network function in response to the first authentication message. Here, the second authentication message comprises a challenge packet and a second identity pseudonym, where the mobile communication

network generates the second identity pseudonym in response to the first authentication message comprising a first identity pseudonym. The processor completes authentication with the mobile communication network using the challenge packet and locally stores the second identity pseudonym.

(213) In certain embodiments, locally storing the second identity pseudonym comprises replacing the first identity pseudonym with the second identity pseudonym. In certain embodiments, the first identity pseudonym and second identity pseudonym are one-time tokens for communicating a permanent subscriber identity of the fourth apparatus in a concealed manner. In certain embodiments, the second network function comprises an AAA server in the mobile communication network, where the AAA server generates the second identity pseudonym in response to the first authentication message comprising a first identity pseudonym. Here, the mobile communication network locally stores the first identity pseudonym and the second identity pseudonym.

(214) In some embodiments, the previous registration with the mobile communication network comprises the processor: sending a registration message to a first network function in the mobile communication network via the 3GPP access network and receiving a first identity pseudonym for the fourth apparatus in response to the registration message comprising the first indicator. In such embodiments, the first authentication message comprising a first indicator and a SUCI for the fourth apparatus, where the first indicator comprises an indication that the fourth apparatus has the capability for access authentication for non-3GPP access in an EPS.

(215) In certain embodiments, the indication that the remote unit has the capability for access authentication for non-3GPP access in an EPS comprises a 5GMM capability information element. In some embodiments, the processor determines whether an identity pseudonym is preconfigured prior to sending the registration message, where the processor includes the first indicator in response to determining that no identity pseudonym is preconfigured.

(216) In some embodiments, a UDM in the mobile communication network generates the first identity pseudonym. In other embodiments, an HSS in the mobile communication network generates the first identity pseudonym.

(217) Disclosed herein is a fourth method for using a pseudonym for access authentication over non-3GPP access, according to embodiments of the disclosure. The fourth method may be performed by a UE, such as the remote unit **105**, the UE **205**, and/or the user equipment apparatus **500**. The fourth method includes sending a first authentication message to a second network function to authenticate with the mobile communication network via the non-3GPP access network. Here, the first authentication message comprises a first identity pseudonym received by the UE during a previous registration with the mobile communication network via the 3GPP access network. The fourth method includes receiving a second authentication message from the second network function in response to the first authentication message. Here, the second authentication message comprises a challenge packet and a second identity pseudonym, where the mobile communication network generates the second identity pseudonym in response to the first authentication message comprising a first identity pseudonym. The fourth method includes completing authentication with the mobile communication network using the challenge packet and locally storing the second identity pseudonym.

(218) In certain embodiments, locally storing the second identity pseudonym comprises replacing the first identity pseudonym with the second identity pseudonym. In certain embodiments, the first identity pseudonym and second identity pseudonym are one-time tokens for communicating a permanent subscriber identity of the UE in a concealed manner. In certain embodiments, the second network function comprises an AAA server in the mobile communication network, where the AAA server generates the second identity pseudonym in response to the first authentication message comprising a first identity pseudonym. Here, the mobile communication network locally stores the first identity pseudonym and the second identity pseudonym.

(219) In some embodiments, the previous registration with the mobile communication network

comprises the UE sending a registration message to a first network function in the mobile communication network via the 3GPP access network and receiving a first identity pseudonym for the UE in response to the registration message comprising the first indicator. In such embodiments, the first authentication message comprising a first indicator and a SUCI for the UE, where the first indicator comprises an indication that the UE has the capability for access authentication for non-3GPP access in an EPS.

(220) In certain embodiments, the indication that the remote unit has the capability for access authentication for non-3GPP access in an EPS comprises a 5GMM capability information element. In some embodiments, the fourth method includes determining whether an identity pseudonym is preconfigured prior to sending the registration message, where the first indicator is included in the registration message in response to determining that no identity pseudonym is preconfigured.

(221) In some embodiments, a UDM in the mobile communication network generates the first identity pseudonym. In other embodiments, an HSS in the mobile communication network generates the first identity pseudonym.

(222) Embodiments may be practiced in other specific forms. The described embodiments are to be considered in all respects only as illustrative and not restrictive. The scope of the invention is, therefore, indicated by the appended claims rather than by the foregoing description. All changes which come within the meaning and range of equivalency of the claims are to be embraced within their scope.

Claims

1. A user equipment (“UE”) comprising: a processor; and a memory coupled to the processor, the processor configured to cause UE to: determine whether an identity pseudonym is preconfigured; send a registration message to a first network function in a mobile communication network via a Third Generation Partnership Project (“3GPP”) access network, the registration message comprising a first indicator and a subscription concealed identifier (“SUCI”) for the UE, wherein the first indicator comprises an indication that the UE has a capability for access authentication for non-3GPP access in an evolved packet system (“EPS”), and wherein the processor is configured to cause the UE to include the first indicator in response to determining that no identity pseudonym is preconfigured; receive a first identity pseudonym for the UE apparatus in response to the registration message comprising the first indicator; and perform access authentication via a non-3GPP access network using the first identity pseudonym.
2. The UE of claim 1, wherein the first identity pseudonym is a one-time token for communicating a subscriber permanent identifier (“SUPI”) of the UE in a concealed manner.
3. The UE of claim 1, wherein the indication comprises a fifth generation mobility management (“5GMM”) capability information element.
4. The UE of claim 1, wherein, to perform access authentication via the non-3GPP access network using the first identity pseudonym, the processor is configured to: send a first authentication message to a second network function to authenticate with the mobile communication network via the non-3GPP access network, the first authentication message comprising the first identity pseudonym; receive a second authentication message from the second network function in response to the first authentication message, the second authentication message comprising a challenge packet and a second identity pseudonym, complete authentication with the mobile communication network using the challenge packet; and locally store the second identity pseudonym.
5. The UE of claim 4, wherein the second network function comprises an authentication, authorization, and accounting (“AAA”) server in the mobile communication network, and wherein, to locally store the second identity pseudonym, the processor is configured to replace the first identity pseudonym with the second identity pseudonym.
6. A unified data management (“UDM”) comprising: a processor; and a memory coupled to the

processor, the processor configured to cause the UDM to: receive a registration request from a user equipment (“UE”), wherein the registration request contains a first indicator and a subscription concealed identifier (“SUCI”) for the UE; acquire an identity pseudonym for the UE in response to receiving the first indicator, wherein the identity pseudonym is usable to authenticate the UE for non-3GPP access in an evolved packet system (“EPS”); store a mapping of the identity pseudonym to a subscriber identity of the UE; initiate a UE parameter update (“UPU”) procedure; and send the identity pseudonym to the UE within UPU data.

7. The UDM of claim 6, wherein the first indicator comprises an indication that the UE has a capability for access authentication for non-3GPP access in an evolved packet system (“EPS”) and a subscriber identity of the UE.

8. The UDM apparatus of claim 7, wherein the indication comprises a fifth generation mobility management (“5GMM”) capability information element.

9. The UDM of claim 6, wherein the identity pseudonym comprises a one-time token for communicating a subscriber permanent identifier (“SUPI”) of the UE in a concealed manner.

10. The UDM of claim 6, wherein, to acquire the identity pseudonym, the processor is configured to create the identity pseudonym using: a subscriber permanent identifier (“SUPI”) of the UE, an international mobile subscriber identity (“IMSI”), or a combination thereof.

11. The UDM of claim 6, wherein, to acquire the identity pseudonym, the processor is configured to: encrypt the subscriber identity, generate a unique value using a random number generator and the subscriber identity, and generate a hash value using a hash function and the subscriber identity, or a combination thereof.

12. The UDM of claim 6, wherein, to acquire the identity pseudonym, the processor is configured to send the subscriber identity to one of: a Home Subscriber Server (“HSS”) or an authentication server in a mobile communication network and receiving the identity pseudonym from the one of a HSS and an authentication server.

13. The UDM of claim 6, wherein, to store the mapping of the identity pseudonym to a subscriber permanent identifier (“SUPI”) of the UE, the processor is configured to send the identity pseudonym to a Home Subscriber Server (“HSS”) in a mobile communication network.

14. The UDM of claim 9, wherein, to send the identity pseudonym to the UE, the processor is configured to: send the identity pseudonym in a registration accept message.

15. An authentication, authorization, and accounting (“AAA”) server comprising: a processor; and a memory coupled to the processor, the processor configured to cause the AAA server to: receive a first authentication message to authenticate a user equipment (“UE”) with a mobile communication network via a non-3GPP access network, the first authentication message comprising a first identity pseudonym for the UE, wherein the first identity pseudonym is received by the UE during a previous registration with the mobile communication network via a Third Generation Partnership Project (“3GPP”) access network; retrieve an authentication vector for the first identity pseudonym; create a second identity pseudonym for the UE; store the second identity pseudonym in the mobile communication network; forward the second identity pseudonym to a home subscriber server (“HSS”); send a second authentication message to the UE, the second authentication message comprising the second identity pseudonym and a challenge packet derived from the authentication vector; and complete authentication with the UE.

16. The AAA server of claim 15, wherein the first identity pseudonym and second identity pseudonym are each one-time tokens for communicating a subscriber permanent identifier (“SUPI”) of the UE in a concealed manner, wherein the first and second identity pseudonyms are mapped to the SUPI of the UE, wherein, to store the second identity pseudonym, the processor is configured to replace the first identity pseudonym with the second identity pseudonym.

17. The AAA server of claim 15, wherein, to retrieve the authentication vector, the processor is configured to cause the AAA server to send the first identity pseudonym to the HSS in the mobile communication network and receive a subscriber permanent identifier (“SUPI”) of the UE, wherein

the HSS stores a mapping of the first identity pseudonym to the SUPI of the UE, and wherein, to store the second identity pseudonym, the processor is configured to cause the AAA server to send the second identity pseudonym to the HSS.

18. The AAA server of claim 15, wherein the second identity pseudonym is created using at least one of: a subscriber permanent identifier of the UE and an international mobile subscriber identity.
 19. The AAA server of claim 15, wherein, to create the second identity pseudonym, the processor is configured to: encrypt a subscriber permanent identifier (“SUPI”) of the UE, generate a unique value using a random number generator and the SUPI of the UE, generate a hash value using a hash function and the SUPI of the UE, or a combination thereof.
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